

Research Paper

An improved particle swarm optimisation algorithm for optimum placement and sizing of biogas-fuelled distributed generators for seasonal loads in a radial distribution network

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ARTICLE INFO

Keywords:

Food waste
Biogas
Distributed generators
Load growth
Fuel cells
Internal combustion engine

ABSTRACT

Optimal allocation is critical for effective planning and operation of grid connected distributed generator(s) (DGs) subject to efficient optimisation approach(es). In this paper, an improved particle swarm optimisation (IPSO) algorithm based on weighted randomised acceleration coefficients and adaptive inertia weight for dynamic movement of the particles towards an optimal solution is proposed. The technique is expected to determine optimal placement, sizing and power factor of biogas-fuelled distributed generators (B-DGs such as internal combustion engine (ICE) and fuel cells (FC)) in a radial distribution network. The proposed IPSO is applied to simultaneously optimise three technical objectives namely total active power loss (TAPL), voltage deviation (VD) and voltage stability index (VSI) considering load growth and seasonal voltage dependence. The proposed IPSO is validated using an IEEE 33 bus radial distribution network to evaluate its effectiveness through various combinations of B-DGs for different scenarios and cases with a maximum of 3 B-DGs. Some of the key findings indicate that operating 3 ICEs at optimal power factor reduces TAPL by 93.52 %, reduces VD by 99.62 % and improves VSI by 23.32 % compared to 61.80 % TAPL reduction, 94.77 % VD reduction and 17.89 % VSI improvement for 3 FCs operating at unity power factor. The proposed IPSO gives better results than the standard PSO in terms of solution quality, convergence speed and statistical results. It is also perceived to be better compared to most of the recent state-of-the-art optimization algorithms found in literature.

1. Introduction

Increase in municipal solid waste (MSW) generation as well as an unprecedented increase in energy demand are two fundamental issues of concern in Africa and other developing countries. The major cause of the duo of MSW proliferation and daunting energy demand could be attributed to soaring population growth, rapid urbanisation and industrialisation with improved economic status and changing consumption

patterns (Alao et al., 2022a). Conventionally, landfilling is the most common method of waste management in developing countries as 90 % of the waste generated in South Africa is landfilled (Sekoai et al., 2023). Waste landfill sites are environmentally and socially unfriendly as a lot of health-related problems occur due to air, water, and land pollution.

Furthermore, coal is the main source of power generation in South Africa contributing about 88.8 % of the total generation (Ayamolowo et al., 2022). The environmental and health risks associated with

Abbreviation: AD, Anaerobic digestion; MSW, Municipal solid waste.; DISCOs, Distribution companies; DGs, Distributed generators; CHP, Combined heat and power; GDP, Gross domestic product; DNO, Distribution network operator; ICE, Internal combustion engine; IEEE, Institute of Electrical and Electronic Engineers; FCs, Fuel cells; PSO, Particle swarm optimisation.; FW, Food waste; IPSO, Improved particle swarm optimisation; MSW, Municipal solid waste; PF, Power factor; HFEO, Hybrid Fuzzy Equilibrium Optimizer HFEO; TSO, Transient Search Optimization; EGWO-PSO, Enhanced grey wolf optimizer and particle swarm optimization; AREP-EGWO-PSO, Available renewable energy potential-based, Hybrid-enhanced grey wolf optimizer–particle swarm optimization; CLTBO, Comprehensive teaching learning-based optimization; MVO, Multiverse optimization; AREP-EGWO-PSO, Hybrid-enhanced grey wolf optimizer–particle swarm optimization; PSO, Particle swarm optimization.; GA, Genetic algorithm; GA/PSO, Hybrid genetic algorithm and particle swarm optimisation and genetic algorithm; SOS-NNA, Symbiosis organism search and neural network algorithm; ALO, Ant lion optimisation; IMOHO, Improved Elephant Herding Optimization; IWO, Invasive weed optimisation; MOPSO, Multi-objective particle swarm optimisation; I-DBEA, improved decomposition based evolutionary algorithm.

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<https://doi.org/10.1016/j.egy.2024.07.049>

Received 16 November 2023; Received in revised form 29 January 2024; Accepted 24 July 2024

Available online 29 July 2024

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coal-based power generation call for adoption of sustainable and environmentally friendly sources for power generation. In a bid to address environmental effects caused by carbon emission due to coal power plant and to ensure increased generation capacity, a long-term energy roadmap was initiated by Department of Energy under the Integrated Resource Plan (IRP) spanning from 2010 to 2030. An integral part of IRP is Renewable Energy Independent Power Provider Procurement Programme (REIPPPP) which allows the independent power producers (IPP) to invest in small scale renewable distributed generations (DGs) which can be integrated to the main grid. The intermittent and weather dependent nature of renewable energy resources such as wind and solar could cause some technical issues on the network. These impacts depend on the capacity and penetration level of the DGs. For a small-scale DG, the impact might not be disastrous on the grid. However, as the capacity and penetration level increase, the effects such as system (grid) instability, rise and fluctuation in voltage level, poor quality power injection and harmonics as well as interruption in the protective (relay) coordination of the existing network can become pronounced. To minimize the severity of these impacts on the South African power grid for example, the national grid code specifies that the bus voltage must be maintained within 0.9 p.u and 1.08 p.u of the nominal value (NERSA, 2022) while a total harmonic distortion value of up to 6 % is permitted for renewable DGs connected to medium voltage (up to 33 kV) distribution network (Thopil et al., 2018).

Unlike other renewables (wind and solar) whose outputs are intermittent, biogas (renewable gas obtained from biomass) can produce firm (stable) power and can be used as a substitute for fossil-fuelled (natural gas or diesel fuel) power plants making it an important renewable energy that needs to be maximally explored. Biogas is produced from the decomposition of organic materials such as food waste in a digester through anaerobic digestion (AD) process. The biogas is fed into an ICE or a gas turbine coupled with an electric generator or into an inverter-based FC for electric power and heat generation in CHP application. The technological advancement in AD technology design has resulted in the production of scalable digesters ranging from small to large-scale. Small-scale digesters are those that can produce biogas for household cooking and lighting usage while large-scale centralized digesters are those that can produce biogas of several thousand cubic meters for CHP application (Vasco-Correa et al., 2017). In the same vein, recent advancement in power generation technologies has led to the proliferation of scalable power engines (such as ICE), FC and micro-gas turbine (MGT) that can directly use biogas as fuel, making B-DG more possible and affordable.

Therefore, for a typical municipality or community, a centralised biogas digester coupled with an appropriate prime mover (electrical/heat conversion engine) can be connected to a distribution network for distributed power generation. The integration of B-DGs to the distribution network has many advantages for the distribution network operators (DNOs), DG owners and the community where the plants are located. The integration of B-DGs can enable increased power transfer capability and regulate system voltage (Barik and Das, 2020) and by extension ensure maximum capacity release, avail the network of being less congested and provide stability to the system. Other benefits are power loss reduction, voltage profile improvement, and improved reliability of power supply as well as deferment in distribution network upgrade investment, reduced operation and maintenance costs, and reduction in energy prices (Ebrahimi et al., 2013). More importantly is the reduction in the emission of greenhouse gases and by extension mitigating the impact of climate change. However, for all these benefits to be maximally achieved and to guarantee economic, environmental as well as technical viability of the system, optimal planning of the B-DGs when connected to the distribution network are very crucial.

1.1. Literature survey

The optimal planning of B-DGs involves finding adequate locations

and their proper sizes (Ali et al., 2021; Sellami et al., 2022). Finding appropriate location and size for DGs are very critical issues as non-optimal planning can subject the system to adverse effects worse than when no DG was connected. While finding appropriate location for a DG, it is important to consider future load growth in the planning scheme. Load growth being a natural phenomenon affects both power loss and voltage profile of any distribution system (Kashyap et al., 2020). It equally affects loading of the feeder which influences the need to upgrade the distribution system. Another essential consideration in the optimal planning of DG involves considering various consumer types such as residential, commercial, and industrial and their seasonal load demands from the onset of the planning scheme.

Various approaches have been applied to optimally determine the size and location of DGs on the distribution or sub-transmission network. Techniques ranging from analytical (Acharya et al., 2006; Viral and Khatod, 2015), mathematical programming (Boljevic, 2014; Yadegari et al., 2020) and artificial intelligence (Ali et al., 2021; Sellami et al., 2022; Hantash et al., 2020; Salimon et al., 2023; Borousan and Hamidan, 2023) considering single or multiple objective functions have been applied. Due to the non-linear, mixed integer, non-convex and multi-objective nature of DGs placement and sizing, the meta-heuristic optimisation algorithms are better used in order to produce acceptable results (Amini and Jalilian, 2021).

Extensive studies using different meta-heuristic optimization algorithms for optimal location and capacity determination of DGs have been presented by various scholars in recent times. In (El-Ela et al., 2021), optimal place and capacity of biomass-based DGs are determined using an adaptive equilibrium optimizer (AEO). The study was aimed at maximising the economic and environmental benefits of biomass-DG while improving the technical performance of the power network. Although load growth was considered in this study but constant power load was applied. Also, seasonal variability of load demand for different consumers were not captured in the study. In a research undertaken by (Kumar and Injeti, 2022), a multi-objective velocity butterfly optimization algorithm (MOVBOA) was adopted for optimal placement of dispatchable distributed generators in on distribution network. In this study, the influence of load growth, consumer types and their seasonal load demands were not considered.

A hybrid of fuzzy and equilibrium equalizer (F-EO) was adopted by (Yehia et al., 2022) to find the optimal location and size of multiple DGs on the distribution network. In their work, fuzzy logic is applied to dynamically adjust the weights of the objective functions. In (Rafi and Dhal, 2020), an hybrid particle swarm optimiser (PSO) and dragon fly (DF) algorithm was used to optimally locate and determine the correct size of a type-1 DG in addition to reconfiguring the distribution network. An improved whale optimisation algorithm (WOA) was proposed by (Prasad et al., 2023) to determine the location and size of multiple DGs on distribution networks based on a multi-objective function. In Ref (Sellami et al., 2022), an MOPSO was applied to determine the location and size of multiple DGs in a distribution network considering multiple objectives. However, in their study, the load demands were assumed constant and no-load growth was considered. In Ref (Raj and Saravanan, 2023), the optimal location and size of DSTATCOM and DG were found by using Dwarf Mongoose Optimization (DMO). In their study, loss sensitivity factor (LSF) was adopted to find candidate buses for the DG and DSTATCOM while the optimisation algorithm was applied to determine the optimal location and size of the DG and DSTATCOM. In all the aforementioned works such as (Sellami et al., 2022; Salimon et al., 2023; Borousan and Hamidan, 2023; Prasad et al., 2023; Raj and Saravanan, 2023), neither was load growth nor the voltage dependent seasonal load model considered individually or simultaneously.

Some recent studies have considered voltage dependent load models in the optimal placement and sizing of DG in distribution networks. In (Nguyen et al., 2021), a hybrid neural network algorithm was employed for optimal allocation of multiple DGs and capacitor banks considering the voltage dependency of load models. However, the yearly growth of

load and seasonal variations in consumers' demands were not captured in their study. Authors in (Kumar et al., 2017) considered load growth as well as voltage dependent load models while determining the optimal location and sizing of different types of DG, however, the seasonal variability of load demands was not put into consideration. In the work of (Mejia et al., 2022), voltage dependent load models was considered for optimum planning of active distribution systems in conjunction with network reconfiguration, but load growth and seasonal load demands were not considered. In another work by (Ahmed et al., 2023), time-varying voltage dependent load models were considered when determining the optimal location and sizing of multiple DGs, the impact of load growth and varied consumption by consumers in the days and nights of the seasons were not captured in their study. In the work of (Essallah and Khedher, 2020), mixed particle swarm optimisation (MPSO) was applied for optimisation of distribution system operation by network reconfiguration and DG integration. In this paper, load growth, voltage dependent loads as well as seasonal load variability were not put into consideration.

Considering the effect of load growth and load models, optimal sizing and location of open-UPQC in distribution networks were undertaken by (Amini and Jalilian, 2021) using genetic algorithm (GA). Similarly, (Hassanzadeh Farda and Alireza Jalilian, 2018) considered the influence of load growth in the optimal location and size determination of DG using PSO. In both studies (Amini and Jalilian, 2021; Hassanzadeh Farda and Alireza Jalilian, 2018) voltage-dependent load models for various consumer types and seasonal load variations were not taken into account.

In an optimal DG allocation problem, operating the equipment at an optimal power factor could guarantee the optimum power generation and minimum loss when connected to the distribution network. Several previous studies have determined optimal location and sizing of DGs using fixed power factors which have been pre-set prior to the optimisation process (Ali et al., 2021; Salimon et al., 2023; Raj and Saravanan, 2023; Mouwafi et al., 2022). Others have used the combined power factor of the connected loads as the optimal power factor of the DGs (Ahmed et al., 2023; Murty and Kumar, 2015). The assumptions in both cases may lead to a non-optimal global result in the optimisation algorithm (Nguyen et al., 2021; Huy et al., 2020). Therefore, finding the optimal operating power factor for DGs when connected to the distribution network should be further investigated.

Another important observation from the literature is that most of the DGs considered are wind and solar (Ahmed et al., 2023; Hassanzadeh Farda and Alireza Jalilian, 2018; Ullah et al., 2020; Doagou-Mojarrad et al., 2013), biomass (El-Ela et al., 2021), natural gas-powered internal combustion engine, micro gas-turbine and fuel cell (Doagou-Mojarrad et al., 2013). Although, natural gas is more environmentally friendly than diesel or gasoline, its combustion still emits some greenhouse gases thereby having environmental and sustainability issues. The use of biogas obtained from FW of MSW of the urban area where the DG plants are located could be an important substitute for natural gas in powering a gas-fired DG.

Among the most frequently used optimisation algorithm for optimal DG planning is PSO. The popularity in the usage of PSO could be attributed to its simplicity, very few parameter selections, and that it does not require any good initial solution to start the algorithm iteration process (Hassanzadeh Farda and Alireza Jalilian, 2018). It can also generate high-quality solutions within less calculation time and has a more stable convergence characteristic than other stochastic methods (Sellami et al., 2022). However, PSO is bedevilled with a major problem of premature convergence and being trapped in local optimum. In a bid to handle these problems, an IPSO is proposed which is capable of ensuring a balance in the exploration and exploitation phases hence, improving the performance for better output.

1.2. Research gap and contributions

Interestingly, the reviewed studies provide important insights into the optimal DG planning problem, to the best of the authors' knowledge none of them has considered a framework incorporating voltage-dependent load models, load growth, demand variability and optimal power factor simultaneously for optimal allocation of biogas-powered DGs in a distribution network. Therefore, to address the shortcomings identified in the reviewed studies, this paper proposes an IPSO for optimal planning of B-DGs in a DN. The main contributions of this paper are:

- A seasonal voltage dependent load model for optimal location and placement of biogas-powered DG is considered.
- Load growth for the available loads in the mentioned load model is considered
- The impact of demand variability during seasonal time-of-use is investigated.
- An IPSO is proposed to find the optimal locations, sizes and power factor of these DG equipment by simultaneously optimising multiple objectives. Various combinations of ICE- and FC-DGs are considered
- The proposed IPSO is tested on a standard radial distribution network (i.e., IEEE 33 bus system)
- The obtained results are compared with the standard PSO and other existing methods in the literature.
- Convergence and statistical analysis are performed to demonstrate the robustness and effectiveness of the proposed algorithm.

Although optimal planning of DG should encompass technical, economic and environmental-based objectives, the technical-based objective is the most important as all other objectives are affiliated with it. For instance, the economic profitability of the DG installation depends largely on total power loss minimisation and how improved the voltage profile is to ensure quality power supply and reliability with enhanced willingness and probably affordability to pay for the energy used by the customers. Based on this premise, this paper is limited to technical-based objectives (i.e., minimising total active power loss, minimising total voltage deviation and maximising total voltage stability index). To solve this multi-objective problem, a weighted sum method is used due to its simplicity, wide acceptability and its ability to avoid the dominance of one objective over another (Mouwafi et al., 2022). Because of the divergent characteristics of the evaluation indices, an index-based approach is applied to normalise the conflicting attributes. This is achieved by taking the ratio of the indices with and without the DG to make each objective unit-less.

2. Biogas-based distributed generator

As previously mentioned, biogas can be used as a substitute for natural gas powered engine, diesel or gasoline-driven machine with little or no modification (Kabeyi and Olanrewaju, 2022; Gupta et al., 2023). Taking a cue from the work conducted by (Alao et al., 2022a), FC and ICE are considered the best and second-best prime movers for biogas-powered DGs due to their complementary environmental, economic and technical advantages. The biogas is produced from AD process using FW in the community where the DGs are to be installed. As the capacity of a DG is inherently limited by the energy resources at any location, it is important to calculate the amount of biogas from the food waste in the location where the DG is to be installed and the electrical power capacity of the DG units (i.e., FC and ICE).

2.1. Biogas production from FW

Biogas is a mixture of methane and carbon dioxide produced from microbial decomposition of organic wastes such as food waste under an anaerobic environment. The volume of biogas (methane) per kilogram

of food waste in a digester could be determined theoretically as follows:

$$V_{CH_4A(t)} = V_{CH_4} \times F_{ORG(t)} (1 - \xi) \quad (1)$$

where, $V_{CH_4A(t)}$ is the actual methane produced, V_{CH_4} is the gross methane extractable from food waste, ξ is the fraction of food waste that does not decompose whose value is taken as 0.15 (Ayodele et al., 2019), $F_{ORG(t)}$ is the amount of food waste available in year t . The detailed process and procedure for determining $V_{CH_4A(t)}$ can be found in Ref (Alao et al., 2022b).

To enhance the energy quality of biogas, the recovered methane is purified by removing impurities such as hydrogen sulphide, and carbon dioxide. Carbon dioxide does not support combustion while hydrogen sulphide gives an obnoxious odour. It also poisons the catalyst in the FC while causing corrosion in the ICE which may lessen the useful lifetime of the equipment. Therefore, the amount of purified methane is obtained using Eq. 2.

$$V_{CH_4P(t)} = V_{CH_4A(t)} \times \nu \quad (2)$$

where ν is the extent of purification whose value is assumed as 87 % (Saur and Milbrandt, 2014).

2.1.1. Power generation from fuel cell

An FC is a device that converts the chemical energy in fuels (usually hydrogen) into direct current (DC) power with zero or near-zero emissions. The main by-product of an FC chemical reaction is water vapour which portends no danger to the environment. FCs are good candidates for DG application due to their scalability, high electrical efficiency, low pollutant emissions and production of high-quality waste heat (Arsalis, 2019). The main fuel for FCs is hydrogen gas which can be obtained through water electrolysis or steam reforming process. The gross electrical power capacity $P_{grossFC(t)}$ of an FC in (kW) can be determined as follows:

$$P_{grossFC(t)} = \frac{\dot{m}_{H_2(t)} \times LHV_{H_2} \times \eta_{comp} \times \eta_{FC}}{\tau} \quad (3)$$

where $\dot{m}_{H_2(t)}$ is the mass of hydrogen extracted from biogas, η_{comp} is the compression efficiency given as

95 % (Lagorse et al., 2008), LHV_{H_2} is the lower heating value of H_2 gas given as 33.3 kWh/kg while the electrical efficiency (η_{FC}) of FC is assumed as 60 % (Ayodele et al., 2019). The parameter τ is operating time (in hours) of FCs and is assumed to be 8760 h. Although, this is an idealistic assumption, the essence is to determine the maximum capacity of the FCs based on the yearly available biogas production.

To avoid repetition, an analytical approach developed by (Alao et al., 2022b) is adopted to determine $\dot{m}_{H_2(t)}$. Since the electrical power generated by FC is direct current (DC), an inverter is required for conversion to alternating current (AC) prior to integration to the distribution network. Therefore, the actual capacity of the FC is given as in Eq. 4.

$$P_{FC(t)} = P_{grossFC(t)} \times \eta_{inverter} \quad (4)$$

The inverter efficiency $\eta_{inverter}$ is assumed as 95 % (Ayodele et al., 2020).

2.1.2. Internal combustion engines (ICE)

An ICE is a synchronous based electrical equipment that can be powered by biogas to generate electrical power. It generates active and reactive powers depending on the operating pf . The active power generated (kW) by an ICE is determined using Eq. 5:

$$P_{ICE(t)} = \frac{V_{CH_4P(t)} \times LHV_{CH_4} \times \eta_{ICE}}{3.6 \times \tau} \quad (5)$$

where, $V_{CH_4P(t)}$ is the purified biogas, η_{ICE} is the electric efficiency of the ICE with a value taken as 40 % (Chowdhury, 2021) after subtracting the self-consumption for the plant. LHV_{CH_4} lower heating value of methane

taken as 37.2 MJ/m³ (Ogunjuyigbe et al., 2017), τ is the operating period of ICE while 3.6 is the conversion factor from MJ to kWh. Based on the operating pf , the reactive power generation capacity (kVar) of the ICE is determined as follows according to Eq. 6 (Zeinalzadeh et al., 2015).

$$Q_{ICE(t)} = P_{ICE(t)} \times \tan(\cos^{-1} pf) \quad (6)$$

The apparent power capacity (kVA) of the ICE is determined as in Eq. 7.

$$S_{ICE(t)} = \sqrt{P_{ICE(t)}^2 + Q_{ICE(t)}^2} \quad (7)$$

3. Load modelling

The modelling of load based on yearly load electric load demand is discussed in this section

3.1. Load growth modelling

Due to population growth, industrial expansion, and economic development, the load demands have risen especially in urban areas. People tend to increase their energy demand when they are economically buoyant and vice versa. Therefore, the electrical load growth rate (a) can be modelled as a function of population growth rate and economic growth rate. Since the economic status of a nation can be measured by its GDP, the value of (a) can be analytically determined according to Eqs. 8 and 9.

$$a = k(b + c) - bc \quad (8)$$

$$k = \begin{cases} 1(b + c) \geq 0 \\ 0 \text{ otherwise} \end{cases} \quad (9)$$

where, k is the load growth rate controlling factor, b and c are the percentage population growth rate and GDP of a particular society or country, respectively. The value of b and c can be positive or negative. Positive b signifies population increase while negative b shows population decline. Depending on the economic status of the country, positive c implies economic prosperity and good income for the energy consumers while negative c means economic recession and poor income for the populace. In this paper, the load growth is determined with the assumption that b and c are positive throughout the operation horizon. Since loads (active and reactive) are connected to the distribution network buses, the yearly active and reactive load growths on load buses are modelled according to Eqs. 10 and 11 (Kumar et al., 2017).

$$p_{Di,t} = p_{D0i}(1 + a)^t \quad (10)$$

$$q_{Di,t} = q_{D0i}(1 + a)^t \quad (11)$$

where, $p_{Di,t}$ and $q_{Di,t}$ are the active and reactive load demands at bus i in year t , p_{D0i} and q_{D0i} are the initial active and reactive load demand at bus i of the distribution network in the reference (base) year (0th year), respectively. The parameter (a) is the load growth rate.

3.2. Voltage dependency load model

Most studies concerning optimal DG placement and sizing on power distribution networks have assumed load demands to have constant active and reactive powers. However, inherent characteristics of practical loads have shown a high dependence of power demand on the system voltage (Nguyen et al., 2021). Therefore, studies on distribution systems based on the constant load model assumptions could not be technically feasible. For a real practical situation, load buses are penetrated with a mixture of different load types such as residential, commercial and industrial whose active and reactive power requirements

are dependent on the supply voltage and frequency of the system (Kumar et al., 2017). Depending on the nature of the area being supplied, different load types may be connected to different load buses or every bus of the system has different load types connected to it. The first approach reflects a well-organised area in which DNO has pre-allocated different buses for different load types. The second approach signifies a more practical situation of power distribution system in which the DNO may not be aware of the load types connected to the distribution network. Since the practical distribution system loads are not completely residential, industrial, and commercial, a mixed load model is considered which comprises only these three load types in which every bus of the system has a mixture of these load types connected to it. Again, loading of the distribution system varies in different seasons (summer and winter day and night) of the year due to some atmospheric conditions such as temperature and humidity. It is expected that decrease in temperature as experienced in the winter day and night will introduce significant increase in power consumption as consumers use their heating devices and other equipment such as geysers to keep them warm.

Therefore, the voltage dependent load model incorporating different load levels in seasons of the year can be mathematically modelled as in Eqs. 12 and 13 (Nguyen et al., 2021).

$$P_{Di,t} = \vartheta \times P_{Di,t} \left(l_{pr} \left(\frac{V_i}{V_0} \right)^{\sigma_r} + l_{pc} \left(\frac{V_i}{V_0} \right)^{\sigma_c} + l_{pin} \left(\frac{V_i}{V_0} \right)^{\sigma_{in}} \right) \quad (12)$$

$$Q_{Di,t} = \vartheta \times Q_{Di,t} \left(m_{qr} \left(\frac{V_i}{V_0} \right)^{\omega_r} + m_{qc} \left(\frac{V_i}{V_0} \right)^{\omega_c} + m_{qin} \left(\frac{V_i}{V_0} \right)^{\omega_{in}} \right) \quad (13)$$

where ϑ represents the load level of the distribution system, $P_{Di,t}$ and $Q_{Di,t}$ are the active and reactive power demand of the bus i in year t , $P_{Di(t)}$ and $Q_{Di(t)}$ are the active and reactive power at operating point i in year t , V_i is the voltage at bus i and V_0 is the voltage at operating point. Based on seasonal load variation and loading conditions of the distribution network, the loading level in each season of the year is discretised into three (i.e., low, base and high) as shown in Table 1. It is assumed that the distribution network is lightly loaded (i.e., low load level) in the summer night, moderately loaded (i.e., at base load level) in the summer day and winter night while it is highly loaded (i.e., at high load level) in the winter day.

According to Eqs. 12 and 13, σ and ω are the active and reactive power exponents of the voltage-dependent load model for constant, residential, commercial and industrial load models with subscript r , c and in respectively. The values of the load power exponents for typical representation of summer and winter day and night of a year are presented in Table 2 (Kumar et al., 2018).

The coefficients l_{pr} , l_{pc} and l_{pin} indicate the weight of active power while m_{qr} , m_{qc} and m_{qin} are for reactive powers of each load type, respectively. These values are assigned by the distribution network operator (utility) or the design engineer based on the active and reactive power consumed by each load type with respect to the actual load demands of the network. Since a real practical load demand consists of a mixture of residential, commercial and industrial, therefore the consumption of active and reactive powers relating to residential, commercial and industrial loads are respectively assumed as 50 %, 30 %, and 20 % of the total load demand. So, the corresponding weight factors of these load types are given as follows $l_{pr} = m_{qr} = 0.5$, $l_{pc} = m_{qc} = 0.3$ and $l_{pin} = m_{qin} = 0.2$. It should be noted that $l_{pr} + l_{pc} + l_{pin} = 1$ and $m_{qr} +$

Table 1
Seasonal load levels.

Season	Loading levels		
	Low load (0.75)	Base load (1.0)	High load (1.25)
Summer	Night	Day	-
Winter	-	Night	Day

Table 2

Seasonal load models and their power exponent value (Kumar et al., 2018).

Season	Load duration	Load models					
		Residential		Commercial		Industrial	
		σ_r	ω_r	σ_c	ω_c	σ_{in}	ω_{in}
Summer	Day	0.72	2.96	1.25	3.50	0.18	6.00
	Night	0.92	4.04	0.99	3.95	0.18	6.00
Winter	Day	1.04	4.19	1.50	3.15	0.18	6.00
	Night	1.30	4.38	1.51	3.40	0.18	6.00

$$m_{pc} + m_{pin} = 1.$$

4. Problem formulation

The objective functions to be optimised including the associated constraints are elucidated in this section.

4.1. Objective functions

In this paper, a multi-objective based optimal placement problem of biogas-based DG has been formulated considering two biogas-powered technologies (i.e., FC and ICE) subject to system constraints based on three technical objective functions.

4.1.1. Objective 1: minimisation of total active power loss

The power loss in a distribution network is a function of the current drawn by the connected loads and the line impedance. Usually, the power loss across the distribution network affects the annual revenue of the DNO (Balu and Mukherjee, 2021) and impacts on the power quality. This objective has to be minimised and is formulated based on the active power loss index as follows (Barik and Das, 2020).

Total active power loss index (TAPLI) is defined as

$$F_1 = \min \left(\frac{TAPLI^{DG}}{TAPLI^0} \right) = \min \left(\frac{\sum_{i=1}^{N_{br}} (I_{i,i+1}^{DG})^2 \times R_{i,i+1}}{\sum_{i=1}^{N_{br}} (I_{i,i+1}^0)^2 \times R_{i,i+1}} \right) \quad (14)$$

where, $\sum_{i=1}^{N_{br}} (I_{i,i+1}^{DG})^2 \times R_{i,i+1}$ and $\sum_{i=1}^{N_{br}} (I_{i,i+1}^0)^2 \times R_{i,i+1}$ are the total active power loss at the line section between bus i and $i + 1$ with and without DGs (i.e., ICE and FC), respectively. $I_{i,i+1}$ is the line current that flows between bus i and $i + 1$ while $R_{i,i+1}$ is the resistance of the line (conductor) connecting bus i and $i + 1$ while N_{br} is the branch number. The value $I_{i,i+1}$ is determined from the load flow algorithm.

4.1.2. Objective 2: minimisation of voltage deviation index (VD)

In practice, there is voltage deviation (VD) at each voltage bus from the nominal voltage because of voltage drop along the radial distribution network due to the line impedance. This effect usually determines the quality of power supply and is mostly pronounced at buses far away from the substation. Therefore, to ensure better quality power and improved voltage profile of the radial distribution network, this effect should be minimised. The objective function is formulated as an index as follows (Mouwafi et al., 2022).

$$F_2 = \min \left(\frac{VD^{DG}}{VD^0} \right) \quad (15)$$

$$VD^{DG} = \sum_{i=1}^{nb} (V_i^{DG} - V_0)^2 \quad (16)$$

$$VD^0 = \sum_{i=1}^{nb} (V_i^0 - V_0)^2 \quad (17)$$

where, V_i^0 (p.u) is the bus voltage when no DG is placed on the distribution network. V_0 (p.u) is the reference voltage value which is taken as 1.0 p.u. V_i^{DG} is the bus voltage after the placement of DG on the network and nb is the bus number.

4.1.3. Objective 3: maximisation of total voltage stability index (VSI)

The stability and security of the distribution system under varying operating and loading conditions are anchored on the capability of maintaining the bus voltages within the statutory limit. To ensure this, the VSI of all the buses must be determined in order to identify the weakest and sensitive buses to voltage collapse and instability. Usually, a bus that has VSI close to zero is more liable to voltage collapse and those having VSI close to unity are viable nodes. Therefore, to ensure the stability of the radial distribution network, VSI must be maximised. This objective function is formulated as an index as follows (Hashem et al., 2021).

$$F_3 = \max \left(\frac{\sum_{i=1}^{nb} VSI_{i+1}^{DG}}{\sum_{i=1}^{nb} VSI_{i+1}^0} \right) \quad (18)$$

$$VSI_{i+1}^0 = |V_i^0|^4 - 4(P_{i+1}X_{i+1} - Q_{i+1}R_{i+1})^2 - (P_{i+1}R_{i+1} - Q_{i+1}X_{i+1})|V_i^0|^2 \quad (19)$$

$$VSI_{i+1}^{DG} = |V_i^{DG}|^4 - 4(P_{i+1}X_{i+1} - Q_{i+1}R_{i+1})^2 - (P_{i+1}R_{i+1} - Q_{i+1}X_{i+1})|V_i^{DG}|^2 \quad (20)$$

where, VSI_{i+1}^0 is the voltage stability index at bus receiving end when DG is not connected and VSI_{i+1}^{DG} is the voltage stability index at the receiving end when DG is connected while $|V_i^0|$ and $|V_i^{DG}|$ are the voltage magnitude at sending end bus i without and with DG, respectively. R_{i+1} and X_{i+1} are the resistance and reactance of the line between the sending and receiving end buses, respectively.

4.1.4. Overall objective function

The overall multi-objective function (MOF) is formulated as a weighted sum of all the objective functions. The total MOF is as shown in Eq. 21.

$$MOF = \min (k_p F_1 + k_v F_2 + k_s F_3^{-1}) \quad (21)$$

k_p , k_v and k_s are the weighting factors of each objective function. The sum of these factors must equal unity (i.e., $k_p + k_v + k_s = 1$). In this paper, k_p , k_v and k_s have been respectively assigned values of 0.4332, 0.2937 and 0.2731 (Alao et al., 2023).

4.2. Constraints

The stated objectives are subjected to both equality and inequality constraints.

4.2.1. Power balance constraint

The algebraic sum of the incoming powers to the distribution system must equal the outgoing powers from the system as presented in Eqs. 22 and 23.

$$P_{(t)}^{grid} + \sum_{n=1}^N \sum_{t=1}^T P_{DGn,t} = \sum_{i=1}^n \sum_{t=1}^T P_{Di,t} + \sum_{\ell=1}^{n\ell} \sum_{t=1}^T P_{L\ell,t} \quad (22)$$

$$Q_{(t)}^{grid} + \sum_{n=1}^N \sum_{t=1}^T Q_{DGn,t} = \sum_{i=1}^n \sum_{t=1}^T Q_{Di,t} + \sum_{\ell=1}^{n\ell} \sum_{t=1}^T Q_{L\ell,t} \quad (23)$$

where $P_{DGn,t}$ and $Q_{DGn,t}$ are respectively the active and reactive power

produced by the DG unit n at year t , $P_{(t)}^{grid}$ and $Q_{(t)}^{grid}$ represent grid active and reactive powers for year t , respectively, $P_{L\ell,t}$ and $Q_{L\ell,t}$ are active and reactive power losses of line ℓ at year t , respectively $P_{Di,t}$ and $Q_{Di,t}$ are active and reactive powers of load demand at bus i for year t .

4.2.2. Voltage limit constraint

An important consideration for allowing the connection of DG units to the distribution network is to ensure compliance with the grid codes which specify the technical requirements to be met at the connection point on the distribution network. The IEEE standard grid code allows a $\pm 5\%$ voltage variation at the connection point. Therefore, the voltage V_i of each bus of the system must be kept within the statutory limit according to Eq. 24

$$V_{\min} \leq V_i \leq V_{\max} \quad (24)$$

where, $V_{\min} = 0.95$ p.u and $V_{\max} = 1.05$ p.u are the lowest and highest allowable values of voltage (Barik and Das, 2020).

4.2.3. DG capacity limit constraint

The active and reactive power injected by each biogas-based DG must fall within their predetermined limit as presented in Eqs. 25 and 26, respectively

$$P_{DG \min} \leq P_{DG} \leq P_{DG \max} \quad (25)$$

$$Q_{DG \min} \leq Q_{DG} \leq Q_{DG \max} \quad (26)$$

where, $P_{DG \min}$ and $P_{DG \max}$ are the minimum and maximum active power generated by the DG while $Q_{DG \min}$ and $Q_{DG \max}$ are the minimum and maximum reactive capacities generated by the DG.

4.2.4. Power factor limit

For effective power delivery of a DG, it should be operated at optimal power factor (pf^{opt}) according to the following Eq. (27)

$$0.8 \leq pf^{opt} \leq 1 \quad (27)$$

4.2.5. Power flow

In power system analysis and studies, load flow is very key. Due to radial nature of a distribution network, forward backward sweep (FBS) load flow is mostly applied in finding the network's technical variables such as power loss, voltage profile etc. This method has proven to be reliable and easy to implement with fast convergence due to elimination of Jacobian matrix found in conventional load flow methods such as Newton Raphson, Gauss seidel and fast decoupled (Parihar and Malik, 2022). The adopted load flow is based on fundamental circuit laws of Kirchhoff's current and voltage, and is implemented based on three matrices namely bus injection to branch current (BIBC), branch current to voltage (BCBV) and distribution load flow (DLF). A detailed explanation of this method can be found in (Ali et al., 2021).

5. Optimisation algorithm

This section describes the proposed improved particle swarm optimisation algorithm.

5.1. The particle swarm optimisation PSO

The PSO is a population-based nature inspired swarm intelligence computational technique proposed by Kennedy and Eberhart in 1995 (Kennedy and Eberhart, 1995) to solve complex optimisation problems. The philosophy behind PSO follows the social and cooperative activities among some creatures such as birds and fishes that search for their food in flocks. This cooperative activity enables the birds or fishes to expend less effort in search of the food as there is information sharing among them during the search process. In PSO, each particle has a position and

it moves with a velocity in a particular direction in the search space. Therefore, there are two vector components that identify i th particle ($i = 1, 2, \dots, N$) i.e., velocity and position.

In a clearer perspective, for a D-dimensional search space, velocity vector is represented as $v_i = (v_{i1}, v_{i2}, v_{i3}, \dots, v_{iD})$, while the position vector $x_i = (x_{i1}, x_{i2}, x_{i3}, \dots, x_{iD})$. The value N represents the particle population number or swarm size. During the search process, each particle updates its position using its previous position and new information about the velocity. During this position update, two best positions (i.e., personal best ($pbest$) and global best ($gbest$)) emerge among the particles. The best position found for a particle i in the search space so far is called $pbest$ while the best position found for all the particles in the swarm is called $gbest$. After identifying these two best values, the particles update their velocities and positions according to the following equations:

$$v_{iD}^{(k+1)} = w \times v_{iD}^{(k)} + c_1 \times r_1 \times (pbest_{iD}^{(k)} - x_{iD}^{(k)}) + c_2 \times r_2 \times (gbest^{(k)} - x_{iD}^{(k)}) \quad (28)$$

$$x_{iD}^{(k+1)} = x_{iD}^{(k)} + v_{iD}^{(k+1)} \quad (29)$$

where, $v_{iD}^{(k)}$ is the velocity of the i th particle at D th dimension in k th iteration, $x_{iD}^{(k)}$ is the previous position of i th particle at D th dimension, w is the inertia weight while r_1 and r_2 are random numbers uniformly generated in the range $[0, 1]$. The particle position is bounded by the upper and lower limits of the decision variables while the velocity is bounded with the range of minimum (v_{min}) and maximum (v_{max}) $[v_{min} v_{max}]$ to reduce the possibility of the particle leaving the search space, where v_{min} is usually taken as a negative fraction (such as -0.1) of v_{max} . The parameters c_1 and c_2 represent the acceleration coefficients (i.e., cognitive and social, respectively).

It is clear from the above equations that the flight process of a particle in every iteration depends on three control parameters such as inertia weight, acceleration coefficient and random numbers. Therefore, performance of PSO is highly dependent on these parameters. The acceleration coefficients influence the movement of the particle towards its $pbest$ and the $gbest$ of the swarm, the two random numbers influence the convergence of a swarm (Tian et al., 2019) while the inertia weight ensures a trade-off between the local search ability (exploitation) and global search ability (exploration) during the optimisation process (Zhou et al., 2011). It is known that PSO suffers from local optima and premature convergence problems therefore adequate selection of these control parameters (inertia weight and acceleration coefficients) can play a significantly role in achieving a global or near global optimal solution.

5.2. Proposed improved particle swarm optimisation (IPSO)

In standard PSO, linearly decreasing inertia weight is applied while the acceleration coefficients are taken as fixed values (i.e., $c_1 = c_2 = 2$) in the initialisation step and are not changed in the subsequent iterations. In this manner, an excessive iterations are needed for convergence to optimal solution (Tsipianitis and Tsompanakis, 2020). It has been argued that in order to accelerate the convergence, the acceleration coefficients may be unequal as a larger value should be assigned to c_2 than c_1 such that the new position of a particle will be closer to the global best position (Alanazi and Alanazi, 2023). The acceleration coefficient could also be selected randomly since randomization can enable the search process to prevent the solution from being stuck in local optima (Fister et al., 2013). In 2002, Ref (Clerc and Kennedy, 2002) introduced a constriction factor ($\kappa = 0.7298$) as a multiplier to the velocity equation in the standard PSO in order to enhance the convergence rate. Based on this premise, an improved PSO (IPSO) is formulated with adaptively decreasing inertia weight and a randomised acceleration coefficient with constriction factor. The proposed IPSO combines an adaptive inertia weight with a randomised acceleration coefficient in order to avoid local optima entrapment and achieve global search

ability.

5.2.1. Adaptive decreasing inertia weight (ADIW)

The approaches for selecting inertia weight (w) fall into linear time varying (Sellami et al., 2022), non-linear time varying (Hantash et al., 2020), and adaptively adjusted methods (Nickabadi et al., 2011). Generally, a larger inertia weight is applied at the beginning of the search process to ensure global exploration and the value is reduced towards the end of search process for local exploitation. However, the success rate of the particle in each iteration is not captured. Therefore, taking a cue from Nickabadi et al (Nickabadi et al., 2011), an adaptive inertia weight which combines the success rate and the decreasing inertia weight for local search is adopted. According to Ref (Nickabadi et al., 2011), high value of success rate implies that there is possibility that the particles have converged to a point that is far from the optimum and the whole particle population is gradually moving towards the optimal solution. However, when there is low success rate, the particles tend to swing within the area of the optimum point with little or no success rate. For a minimisation problem, the rate of success of the i th particle at the k th iteration is expressed as Eq. 30.

$$\theta_i(k) = \begin{cases} 1, & f(pbest_i(k)) < f(pbest_i(k-1)) \\ 0, & f(pbest_i(k)) = f(pbest_i(k-1)) \end{cases} \quad (30)$$

where $pbest_i(k)$ is the personal best position of the i th particle so far until the k th generation, $pbest_i(k-1)$ the previous personal best position until $pbest_i(k)$ is found, $f(\cdot)$ is the fitness of the objective function being optimised. Hence, the success rate of the entire population is given by Eq. 31.

$$\psi(k) = k_{max}^{-1} \times \sum_{i=1}^n \theta_i(k) \quad (31)$$

where, k_{max} is the maximum value of the iteration, n is the population size, and $\psi(k) \in [0, 1]$ is the ratio of the number of particles which have produced better fitness values in the previous generation. According to the success rate, the inertia weight is linearly and adaptively changing as the iteration progresses according to Eq. 32.

$$w_{new}(k) = w_{max} - (\psi(k) \times (w_{max} - w_{min})) \quad (32)$$

where, w_{max} and w_{min} are the maximum and minimum inertia weights, k and k_{max} are the current iteration and maximum iteration, respectively.

5.2.2. Random acceleration co-efficient

In this paper, a randomised-weighted acceleration coefficient similar to the randomisation parameter in firefly algorithm is applied. The proposed weighted randomised acceleration coefficients are presented according to Eq. 33.

$$c_{1new} = \alpha_1 \times (r_3(0, 1) - 0.5) \quad (33)$$

$$c_{2new} = \alpha_2 \times (r_4(0, 1) - 0.5) \quad (34)$$

Where, α_1 and α_2 are respectively non-negative randomisation weights while r_3 and r_4 are two independent uniformly generated random numbers in the range of $[0, 1]$.

Therefore, the new equation for velocity and position updates are shown in the following equations:

$$v_{iD}^{(k+1)} = \kappa \times [w_{new} \times v_{iD}^{(k)} + c_{1new} \times r_1 \times (pbest_{iD}^{(k)} - x_{iD}^{(k)}) + c_{2new} \times r_2 \times (gbest^{(k)} - x_{iD}^{(k)})] \quad (35)$$

$$x_{iD}^{(k+1)} = x_{iD}^{(k)} + v_{iD}^{(k+1)} \quad (36)$$

6. Application of the proposed method to a case study

Due to the mixed integer nature of the problem, size ($P_{DG(i)}$), location ($loc_{DG(i)}$) and optimal power factor ($pf_{DG(i)}$) of the DGs are initially randomly generated within their upper and lower limits according to the following equations:

$$P_{DG(i)} = P_{DG(i)}^{\min} + rand(P_{DG(i)}^{\max} - P_{DG(i)}^{\min}) \quad (37)$$

$$loc_{DG(i)} = round(loc_{DG(i)}^{\min} + rand(loc_{DG(i)}^{\max} - loc_{DG(i)}^{\min})) \quad (38)$$

$$pf_{DG(i)} = pf_{DG(i)}^{\min} + rand(pf_{DG(i)}^{\max} - pf_{DG(i)}^{\min}) \quad (39)$$

A vector comprising these three parameters will be formed according to

$$X = [P_{DG1}, P_{DG2}, \dots, P_{DG(nDG)}, loc_{DG1}, loc_{DG2}, \dots, loc_{DG(nDG)}, pf_{DG1}, pf_{DG2}, \dots, pf_{DG(nDG)}] \quad (40)$$

Fig. 1 shows the multi-functional flowchart for the IPSO implementation for optimal allocation of B-DGs.

6.1. Case study

The proposed IPSO was applied to determine the optimal location and capacity of FC and ICE for a small-scale distribution network considering a typical township in South Africa called Soweto. The considered township is located within the city of Johannesburg with 43 % of the population of Johannesburg (Ayeleru et al., 2021). The huge population of this township is a plus for its food waste potential which is the feedstock for biogas production. Of note is the proximity of Soweto to one of the four functional landfill sites (LSs) in Johannesburg which is an important factor for setting up a biogas plant facility and a major element for selecting Soweto as the case study in this paper. Table 3 shows the data for determining the biogas and electric capacity of FC and ICE units for the considered case study.

The IEEE 33-bus distribution system is employed in this work to test the applicability of the proposed method. Although, real network of Soweto would have been more appropriate to be used, however due to a lack of detailed network data which could serve the purpose of this study, IEEE 33 bus system (as shown in Fig. 2) being an established universal distribution system for any location is employed. Due to the metropolitan nature of Soweto, the load demands are modelled as a mixture of residential, commercial, and small-scale industrial loads. The active and reactive load exponents presented in Table 2 are defined for both summer and winter seasonal load models

In this system, bus 1 is considered as a slack (reference) bus from where the external grid is connected, and the rest buses are considered as candidate buses for DG integration. The base voltage of this network is 12.66 kV and base apparent power is 100MVA. The technical data of this network can be found in (Aman et al., 2014). The proposed algorithm was implemented in self written codes in MATLAB 2021a software and the simulation is done on Intel core i5–4200 U 1.6 GHz 8 GB RAM laptop computer.

6.2. Assumptions and limitation for this study

The following assumptions and limitations are stated for this paper:

- 1) The DGs are implemented mathematically as a negative load and constant power factor models.
- 2) The case when the feeder requires to be upgraded to accommodate new DG is not included in the study.
- 3) DGs are installed at the PQ buses for load flow purposes.
- 4) The ICE is operated at an optimal power factor while FC is operated at unity power factor
- 5) Maximum of three DGs are connected to the distribution system.

- 6) Since the primary fuel (biogas) for both FC and ICE is obtained from the same source (i.e., food waste), to ensure maximum utilisation of the fuel, 33.33 % of the fuel is utilised by each DG. In this case, the active capacities of the DGs are limited to

$$100 \leq P_{DG} \leq 0.333 \times P_{DG \max} \text{ (kW)} \quad (41)$$

- 7) The parameters for both IPSO and standard PSO are shown in Table 4.

7. Simulation results and discussion

The results for the biogas and DG (ICE and FCs) capacity for the selected study area as well as the impact of integrating these DGs on the distribution network are presented. The impact on the distribution network is manifested as multi-objective such that the overall objective function is minimised. The voltage profiles, total active power losses, voltage stability and voltage deviation indices before and after optimal integration of the DGs are observed for constant and seasonal voltage dependent practical mix load models.

7.1. DG electric capacity potential determination

Using data presented in Table 3 and applying Eqs. 1–7, the load growth, the FW potential, biogas potential and electric power capacities of FC and ICE over the operating time horizon of five years for Soweto townships are estimated and the results are presented in Table 5.

It is observed from Table 5 that both DGs have the potential capacities to feed the load buses of the selected network. Since the maximum active power capacity of each DG is comparable to the total active power demand (i.e., 3715 kW) of the selected network.

7.2. Performance analysis of the DG combination for constant and mixed load model

In this section, two categories are investigated. Each category has two scenarios. Scenario 1 is the base case (i.e., no DG) while scenario 2 has four cases of DG combinations. The first category deals with constant load model while the second considers the seasonal mixed load model for summer and winter day and night under varying load levels.

7.2.1. Constant load model

Category 1: In this case, the performance of the system before and after the installation of FCs and ICE-based DGs to the distribution network under constant load models with an estimated load growth of 4.83 % over a five-year operation horizon is investigated.

Scenario 1: Case 0 (Base case): This is the base scenario before optimisation with no DG connected to the distribution network. The performance metrics of the system over an operating time of five years are displayed in Table 6.

At the base year when no load growth had occurred, the total active and reactive demands of the system are 3.7150 MW and 2.3000 MVar, respectively. With load growth at an estimated rate of 4.83 %, the total active power demand grows from 3.715 MW to 4.7033 MW and the reactive load power increases from 2.3000 MVar to 2.7777 MVar from base year to the final year. Due to load growth, the total active power loss of the system increases tremendously from 202.6771 kW to 339.2824 kW and the system bus voltage drops substantially from the initial 0.9130563–0.8872597 p.u at bus 18. The total VSI and minimum VSI reduces from 25.8647 p.u to 24.2965 p.u and 0.695112–0.6198939 p.u, respectively while VDI increases from 0.1171 p.u to 0.1966 p.u. The implication of the above is that increase in load demand increases the current flow in the network and equally deteriorates (worsens) the performance of the network with poor quality power as exemplified by degraded voltage profile and voltage stability index as well as increased

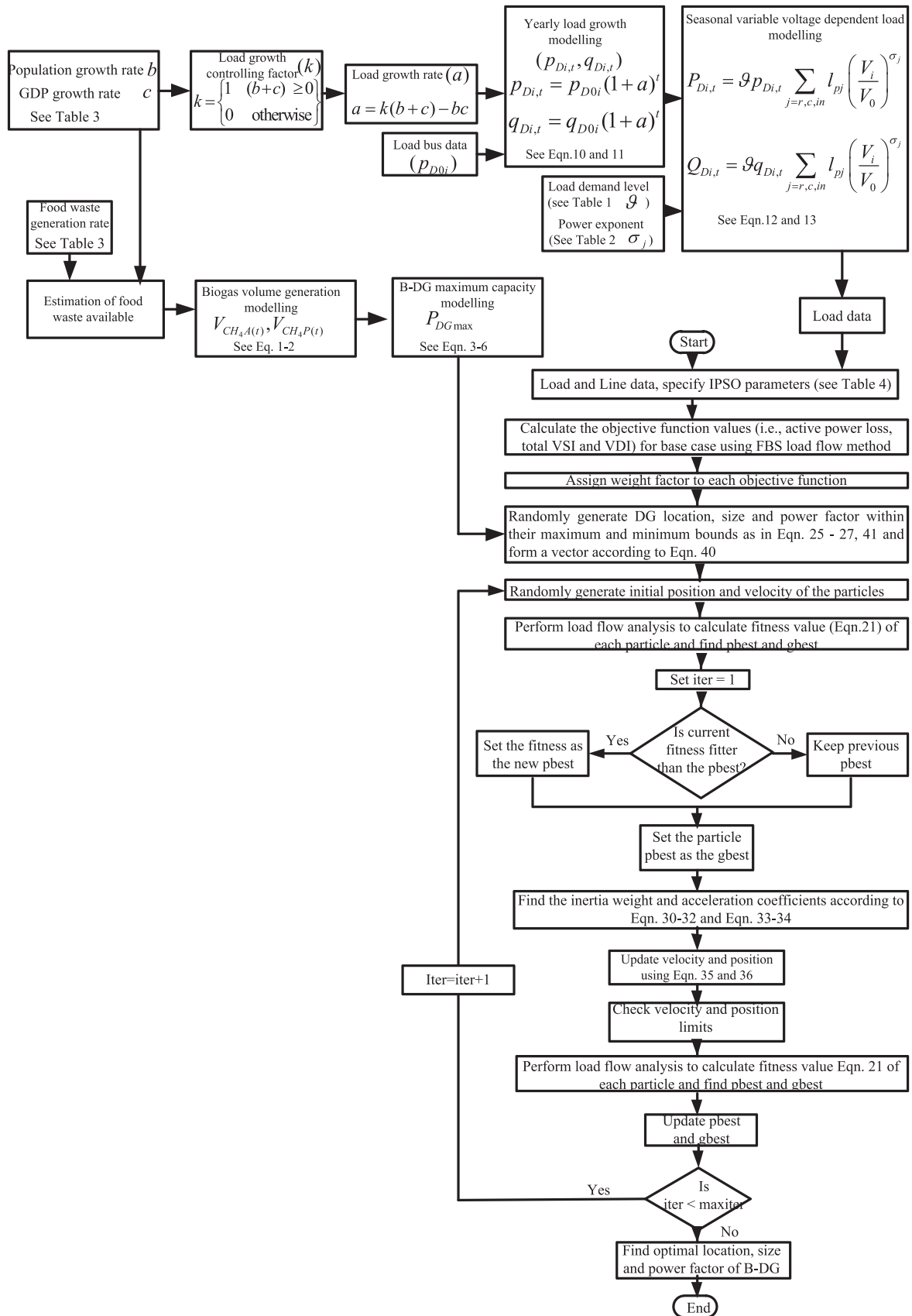
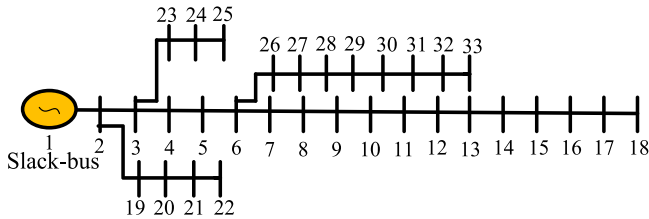


Fig. 1. Multi-functional flowchart of the proposed IPSO for optimal allocation of B-DG.

Table 3

Data for calculating the biogas generation potential of the case study.

Population size	Population growth rate	GDP growth rate	Per capita food waste (kg/capita/week)	Waste collection rate
2.487 million	2.63 %	2.26 %	0.22	69 %
(Alao et al., 2022b)	(Alao et al., 2022b)	(Alao et al., 2022b)	(Oelofse et al., 2018)	(Alao et al., 2022b)

**Fig. 2.** IEEE 33 bus radial distribution system.**Table 4**

The parameters for the IPSO and PSO.

Methods	Population size	max iter	w_{min}	w_{max}	α_1	α_2
IPSO	50	100	0.4	0.9	1.5	2.5
PSO	50	100	0.4	0.9	-	-

voltage deviation from the acceptable limit. Since the line parameters (i. e., resistance and reactance) remain the same, the ohmic power loss (I^2R) increases over the years leading to increase in the active power loss. Active power loss is the major cause of revenue loss to the DNO. However, the integration of DGs can adequately ameliorate this issue which is investigated in the next scenarios and cases.

Scenario 2: This scenario has four cases in which different configurations (combinations) of FC and ICE are integrated to the distribution network for constant load model for concurrent optimisation of three technical objectives. Therefore, considering various DG configurations in the four cases, the results of the optimisation are presented in Table A1 (see the appendix).

For Case 1, allocation of 2 FC and 1 ICE causes the total active power loss to reduce from 202.6771 kW (without DG) in the base year to 28.705831 kW resulting in about 85.84 % power loss reduction. The

total VSI also increases from 25.8647 p.u (without DGs) to 31.078882 p.u in the base year of operation which translates to about 20.16 %. The minimum VSI appreciates from 0.695112 to 0.9539507 p.u which is equivalent to 37.24 % improvement. The VD also recorded some improvement after the integration of the DGs. The VD reduces from 0.1171 p.u in the initial year to 0.00205050 p.u which amounts to about 98.25 % improvement. The system voltage profile also improves with the minimum voltage increased to 0.988201 p.u in the base year in comparison to 0.9130563 p.u before the integration of the DGs. Similar improvement is recorded for other years of system operation.

For Case 2, simultaneous allocation of 1 FC and 2 ICE causes the total active power loss to reduce from 202.6771 kW (without DG) in the base year to 17.025396 kW resulting in about 91.60 % power loss reduction. The total VSI and the minimum VSI also increases from 25.8647 and 0.695112 p.u (without DGs) to 31.631752 p.u and 0.9754334 p.u, respectively in the initial year of operation which translates to about 22.30 % and 40.33 %, respectively. The VD also shows some appreciation after the integration of the DGs. The VD also shows some appreciation after the integration of the DGs as it reduces from 0.1171 p.u in the initial year to 0.00055296 p.u which amounts to about 99.53 % reduction. The system voltage profile also improves with the minimum voltage increasing to 0.993777 p.u in the base year in comparison to 0.91305 p.u before the integration of the DGs. Similar improvement is recorded for other years of system operation.

The concurrent allocation of 3 FCs leads to the total power loss of 77.426047 kW in the base year contrary to 202.6771 kW without the integration of DGs which translates to 61.80 % loss reduction. The system VD and total VSI also recorded some improvement. The VD of the system reduces from 0.1171 p.u in the initial year to 0.00612932 p.u which amounts to about 94.77 % reduction. On the other hand, the total VSI and minimum VSI of the system also improves from 25.8647 and 0.695112 p.u to 30.493037 and 0.9200119 p.u, respectively in the base year of system operation. The minimum voltage of the system increases to 0.978535 p.u in the base year in comparison to 0.91305 p.u prior to the integration of the DGs.

In the case of simultaneous integration of three ICEs, the total power loss reduces from 202.6771 kW to 13.1381206 kW in the base year and from 339.2824 kW to 21.0367506 kW in the fifth year which translates to about 93.52 – 93.80 % loss reduction over a period of five years. Likewise for the system's VD, it reduces from 0.1171 p.u in the base year to 0.0004432060 p.u which amounts to about 99.62 %. Over the operating years, about 99.62 – 99.65 % VD reduction can be achieved. On the other hand, the total VSI of the system also improves from 25.8647 p.u to 31.668805 p.u in the base year of system operation and from 24.2965 p.u to 31.582181 p.u in the fifth year. The total VSI of the

Table 5

Estimated food waste, biogas potential and power generated by FC and ICE for Soweto.

Year	Base	1	2	3	4	5
FW potential (kilotons/annum)	19.6289	20.6005	21.6201	22.6901	23.8132	24.9918
Biogas potential ($\times 10^6$ m ³ /annum)	6.9094	7.2514	7.6103	7.9870	8.3823	8.7972
Biogas DG type	Maximum power capacity of each DG per year ($P_{DG\ max}$) (kW)					
ICE (P_{ICE})	3123.64	3278.24	3440.49	3610.78	3789.49	3977.05
FC (P_{FC})	3573.25	3750.11	3935.72	4130.51	4334.95	4549.51

Table 6

Results for load demand and system performance parameters for constant load without DGs.

Year	$P_{D(t)}$ (MW)	$Q_{D(t)}$ (MVar)	$TAPL^0$ (kW)	sumVSI ⁰ (pu)	VD ⁰ p.u	Min(V) p.u (bus)	MinVSI p.u
Base	3.7150	2.3000	202.6771	25.8647	0.1171	0.9130563 (18)	0.695112
1	3.8945	2.4111	224.4407	25.5780	0.1298	0.9084717 (18)	0.681268
2	4.0826	2.5276	248.6612	25.2785	0.1439	0.9036196 (18)	0.666843
3	4.2798	2.6497	275.6390	24.9654	0.1596	0.8984812 (18)	0.651818
4	4.4865	2.7777	305.7161	24.6383	0.1771	0.8930355 (18)	0.636175
5	4.7033	2.9118	339.2824	24.2965	0.1966	0.8872597 (18)	0.619894

system improves by 23.34–29.99 % over the planning period of five years.

With the location of these DGs at the stated buses in Table A1, the system voltage profile improves substantially in comparison to the base case. To show the influence of these DGs allocation on voltage profile as well as branch currents of the distribution network, the voltage profile and the branches current for each case for the fifth year is presented since this is the year when maximum load is connected to the system and maximum voltage drop occurs. Figs. 3 and 4 show the graphical presentation of the voltage profile and the branches current, respectively.

From Figs. 4 and 5, it is vivid that after integration of FC and ICE in various configurations the voltage magnitudes at all buses improve to higher value while the branches current reduces compared to the base case.

In general, for the investigated scenarios under the constant load model, the allocation of 3 ICEs operating at optimum power factor produces the best optimum results by substantially reducing the power loss and branches current while better improving voltage profile and other network performance parameters. This could be attributed to the fact that DGs that have the capability of supplying active and reactive powers could better enhance the network performance. Followed by this is the integration of 2 ICE and 1 FC.

Again, it is seen that the capacities of the DGs are increasing as the load grows over the years, but their locations remain practically unchanged for each case of DG combinations. This is because, as the load increases the capacity of the DG is expected to increase to be able to meet up with the load increase. It could be said that load growth can only affect the capacity of DGs but may not necessarily influence their locations but rather the DG configuration (hybridization/combination of DGs) can have impact on the location of DGs in the distribution network.

7.2.2. Comparison with existing models

To show the effectiveness of the proposed IPSO, the results obtained for 3 FC (type 1 DG) and 3 ICE (type 3 DG) configurations for the base year (no load growth) are compared with those of standard PSO and those reported by other existing techniques found in literature. The comparison results are tabulated in Table 7. For type 1 DG (such as FC) and operates at unity power factor (UPF), it is observed from Table 7 that the proposed IPSO gives a better result than I-DBEA (Ali et al., 2021), IMOHO (Meena et al., 2018), MOPSO (Kamel et al., 2020), SOS-NNA (Nguyen et al., 2021), CTLBO (Quadri et al., 2018), HFEO (Yehia et al., 2022), TSO (Bhadoriya and Gupta, 2022), GA (Moradi and

Abedini, 2012), PSO (Moradi and Abedini, 2012), GA/PSO (Moradi and Abedini, 2012), and IWO in terms of active power reduction. It performs better than GA (Moradi and Abedini, 2012), PSO (Moradi and Abedini, 2012) and GA/PSO (Moradi and Abedini, 2012) in VD minimisation. The proposed IPSO outweighs the (EGWO-PSO) (Venkatesan et al., 2021a) and AREP-EGWO-PSO (Venkatesan et al., 2021b) in both VD minimisation and VSI maximisation. It also gives better results than standard PSO in all the compared objectives. This shows that the proposed IPSO outperforms the standard PSO and other comparing methods in terms of the solution quality. The outstanding performance of IPSO in comparison with standard PSO could be attributed to the randomised acceleration coefficient and an adaptive inertia weights which dynamically ensure a balance in exploration and exploitation phases and movement toward global solution rather than fixed value used in standard PSO. It should be pointed out that the total DG capacities reported by some of the referenced techniques such as I-DBEA (Ali et al., 2021), IMOHO (Meena et al., 2018), MOPSO (Kamel et al., 2020), SOS-NNA (Nguyen et al., 2021), CTLBO (Quadri et al., 2018) and TSO (Bhadoriya and Gupta, 2022) are higher than the total active demand power (i.e., 3.715 MW) of IEEE 33 bus radial distribution network, hence a violation of the DG capacity constraint which can cause reverse power flow and a total disruption in the system stability. This is a violation of the capacity constraint limit which could result in technical (reverse power flow) and economic (i.e., exorbitant capital and maintenance costs) implications.

In the case of type 3 DG (such as ICE) which operates at optimal PF (OPF), it is shown in Table 8 that the proposed IPSO achieves a lower total active power loss of 13.1381 kW than 14.554 kW in SOS-NNA (Nguyen et al., 2021) and 14.940 kW in TSO (Bhadoriya and Gupta, 2022) whereas it is higher than 11.74 kW and 11.75 kW reported in ALO (Sultan et al., 2023) and MVO (Sultan et al., 2023), respectively. In terms of VD minimisation, the proposed IPSO returns a better output of 0.000443 p.u. than 0.000461 p.u. in ALO (Sultan et al., 2023) and 0.000519 in MVO (Sultan et al., 2023). However, the reported VD 0.000315 p.u. in SOS-NNA (Nguyen et al., 2021), 0.000300 in TSO (Bhadoriya and Gupta, 2022) and 0.0002 in I-DBEA (Ali et al., 2021) are better than that achieved by the proposed IPSO. In terms of VSI maximisation, the result obtained by proposed IPSO (i.e., 0.9775) is better than 0.9774 in SOS-NNA (Nguyen et al., 2021), 0.9767 in TSO (Bhadoriya and Gupta, 2022) and 0.9733 p.u. in I-DBEA (Ali et al., 2021) but worse than 0.9805 in ALO (Sultan et al., 2023) and 0.9797 in MVO (Sultan et al., 2023). By and large, it could be inferred that the proposed IPSO has its own merits as it presents better results than TSO (Bhadoriya

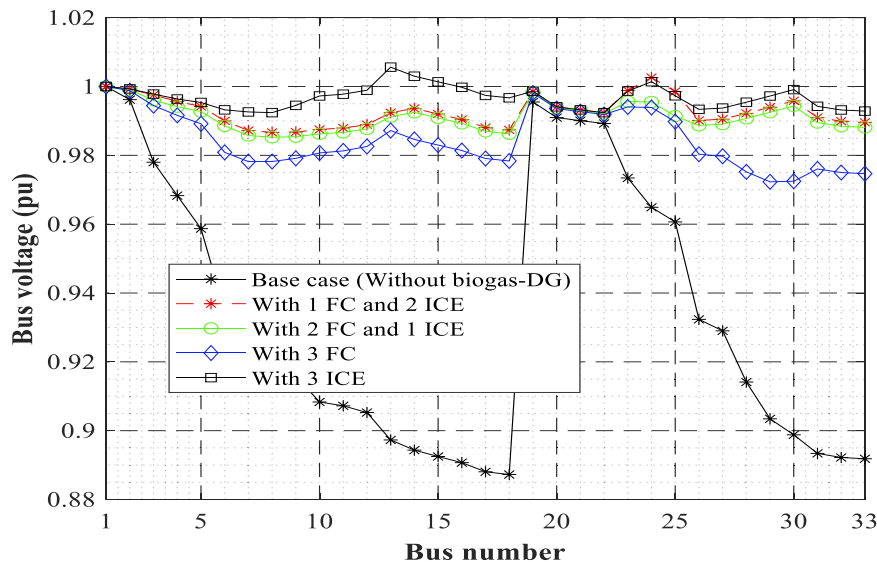


Fig. 3. Voltage profile of fifth year for all the investigated cases.

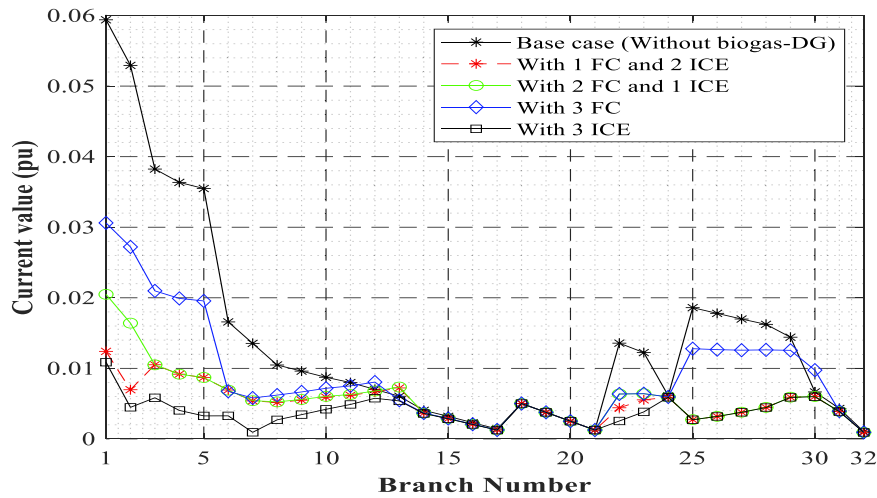
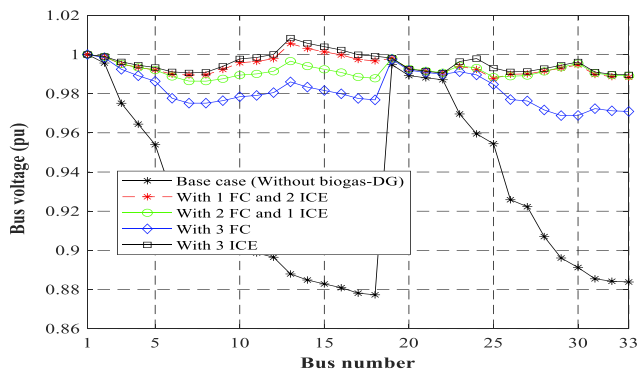
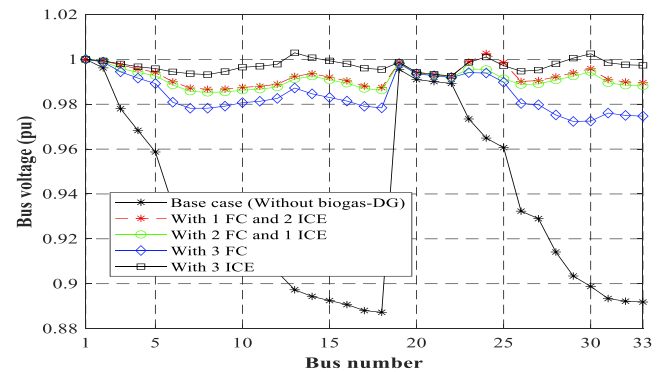


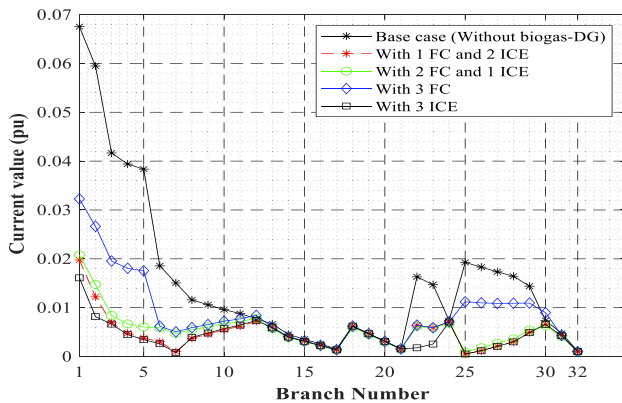
Fig. 4. Branch current of fifth year for all the investigated cases.



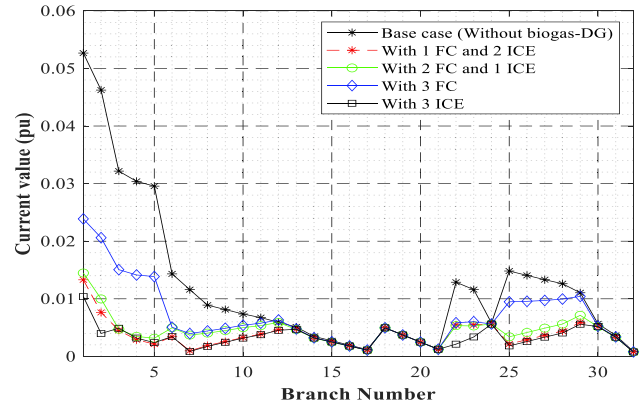
a) Voltage profile for winter day



b) Voltage profile for winter night



c) Branch current for winter day



d) Branch current for winter night

Fig. 5. Voltage profile and branch currents for winter day and night.

and Gupta, 2022), SOS-NNA (Nguyen et al., 2021) and I-DBEA (Ali et al., 2021) in two out of the three compared objective but in one objective than ALO (Sultan et al., 2023) and MVO (Sultan et al., 2023).

7.3. Mixed load model

Category 2: The performance of the system before and after the installation of FC and ICE-based DGs to the distribution network considering different load conditions for seasonal mixed voltage dependent load models with an estimated load growth of 4.83 % over a

five-year operation horizon is presented in this section.

Scenario 1: Case 0 (Base case)

This is the case when no DG (FC or ICE) is connected to the system for three discretised load levels (i.e., low load for summer night, base load for summer day and winter night, and high load for winter day). It should be noted that the day and night referred to in this study represent typical day and night of a year in the seasons concerned. According to Table 9, at the base year when no load growth occurred, the total active and reactive power demands of the system in the summer day are 3.5665 MW and 1.8597 MVar, respectively while at the night when the

Table 7

Comparative results of DG allocation for simultaneous optimisation of real power loss, voltage deviation and voltage stability index for DG operating at unity power factor.

Method	Location (bus No)	DG capacity (MW)	PF	Total DG size (MW)	TAPL (kW)	Loss reduction (%)	VDI (p.u)	Min VSI (p. u)
Base case	-	-	-	-	202.6771	0.000	0.117100	0.6951
I-DBEA (Ali et al., 2021)	13, 24, 30	1.0980, 1.0970, 1.7150	Unity	3.9131	94.8514	53.21	0.000700	0.9650
IMOEHO (Meena et al., 2018)	14, 24, 30	1.0570, 1.0540, 1.7410	Unity	3.8520	95.0000	54.98	0.000800	0.9678
MOPSO (Kamel et al., 2020)	3, 14, 31	1.6762, 0.9553, 1.3013	Unity	3.9328	87.3000	56.93	0.015500	NR
SOS-NNA (Nguyen et al., 2021)	16, 24, 33	1.1234, 1.4532, 1.1386	Unity	3.7152	95.9900	54.59	0.003395	0.9268
CTLBO (Quadri et al., 2018)	13, 24, 30	1.0364, 1.1630, 1.5217	Unity	3.7211	85.9595	59.26	0.002600	0.9481
HFE0 (Yehia et al., 2022)	24, 30, 13	1.2090, 1.2572, 1.0431	Unity	3.5093	77.6570	61.68	NR	NR
TSO (Bhadoriya and Gupta, 2022)	12, 24, 30	1.3372, 1.1852, 1.3368	Unity	3.8592	89.3100	57.67	0.00200	0.9416
IWO (Prabha and Jayabarath, 2016)	14, 18, 32	0.6247, 0.1049, 1.0560	Unity	1.7856	85.8600	57.47	NR	NR
GA (Moradi and Abedini, 2012)	11, 29, 30	1.5000, 0.4230, 1.0710	Unity	2.9940	106.300	-	0.04070	0.9497
PSO (Moradi and Abedini, 2012)	8, 13, 32	1.1770, 0.9820, 0.8290	Unity	2.9880	105.300	-	0.03350	0.9256
GA/PSO (Moradi and Abedini, 2012)	11, 16, 32	0.9250, 0.8630, 1.2000	Unity	2.9880	103.400	-	0.01240	0.9508
EGWO-PSO (Venkatesan et al., 2021a)	14, 24, 30	0.7540, 1.0990, 1.0710	Unity	2.9240	71.4570	64.74	0.01350	0.8813
AREP-EGWO-PSO (Venkatesan et al., 2021b)	13, 24, 31	0.8230, 1.1210, 0.9340	Unity	2.8780	72.5300	64.21	0.01400	0.8871
Standard PSO	13, 24, 31	1.0326 , 1.1304, 1.1404	Unity	3.3034	77.4660	61.78	0.006130	0.9200
Proposed IPSO	13, 24, 30	1.0326 , 1.1404 , 1.1404	Unity	3.3134	77.4260	61.80	0.006129	0.9200

Table 8

Comparative results of DG allocation for simultaneous optimisation of real power loss, voltage deviation and voltage stability index for DG operating at optimal power factor.

Method	Location	DG capacity (kW) (optimal PF)	TAPL (kW)	Loss reduction (%)	VDI (p.u)	Min VSI (p.u)
SOS-NNA (Nguyen et al., 2021)	13, 24, 30	817.00 (0.880), 1433.20 (0.905), 1116.20 (0.725)	14.5540	93.10	0.000315	0.9774
TSO (Bhadoriya and Gupta, 2022)	12, 24, 30	932.13 (0.850), 1053.96 (0.840), 1140.43 (0.820)	14.9400	92.92	0.000300	0.9767
ALO (Sultan et al., 2023)	13, 24, 30	876.15 (0.900), 1190.88 (0.900), 1131.17 (0.710)	11.7400	94.42	0.000461	0.9805
MVO (Sultan et al., 2023)	13, 24, 30	868.38 (0.880), 1198.93 (0.910), 1142.60 (0.710)	11.7500	94.43	0.000519	0.9797
I-DBEA (Ali et al., 2021)	13, 24, 30	749.10 (0.850), 1042.10 (0.850), 1239.50 (0.850)	14.5700	92.81	0.000200	0.9733
Standard PSO	13, 24, 30	828.78 (0.8709), 1078.32 (0.8751), 1080.32 (0.8000)	13.2325	93.47	0.000447	0.9761
Proposed IPSO	13, 24, 30	832.82 (0.8724), 1080.42 (0.8815), 1080.50 (0.8000)	13.1381	93.52	0.000443	0.9775

system is lightly loaded, the total active and reactive power demands are 2.6716 MW and 1.3416 MVar, respectively. For a typical winter day when the demand is high, the total active and reactive power demands in the initial year are respectively 4.4038 MW and 2.2581 MVar whereas at night the total active and reactive demands are reduced to 3.4987 MW and 1.7889 MVar, respectively. With the load growth, the total annual active and reactive powers for both summer and winter day and night and their impacts on the network parameters are depicted in Table 9.

It is observed from Table 9 that in summer and winter day and night, the system parameters are differently affected with the highest impact noticed in the winter day due to high energy demand.

To ameliorate this impact and ensure reliable power supply, the integration of DGs of various combinations is expected to play an important role as will be presented in the following section. It is worth noting that only high loading condition of winter season is considered. This assumption is based on the premise that if the network equipment such as lines and transformers are designed and selected based on the high loading condition experienced in the winter season, they can definitely work perfectly during other seasons when the network is subjected to less loads. However, this assumption is not static or

compulsory as the proposed planning model has the capability to incorporate different loading levels over the planning horizon.

Scenario 2: In this scenario, the impact of the integration of different combinations of DGs (i.e., FCs and ICEs) to the distribution network during the winter day and night are investigated on a case by case basis. The results of optimisation are shown in Table A2 (see Appendix). In winter day for case 1, the allocation of 2 FC and 1 ICE causes the total active power loss to reduce from 243.2314 kW (without DGs) to 29.8459809 kW in the base year of operation which translates to about 87.73 % power loss reduction. The total VSI grows from 25.36686 p.u (without DGs) to 30.96303313 p.u while total VD reduces to 0.002476436 p.u in comparison to 0.138073 p.u before the integration of DGs which amounts to 98.21 % reduction. The system voltage profile also improves with the minimum voltage increasing to 0.9874869 p.u in bus 7 in the base year in comparison to 0.905641 p.u before the integration of the DGs. However, in the night, the system is at base load and the active power loss drops to 17.5369031 kW when compared with 29.8459809 kW active power loss in the daytime. This is equivalent to about 41.24 % loss reduction when compared with daytime. The system voltage profile also improves with the minimum voltage increased to 0.9940593 p.u in the base year in comparison to 0.9874869 p.u in the

Table 9

Yearly seasonal load demand and system performance metrics for mixed load model under varying load levels without DG.

Season	Year	$P_{D(i)}$ (MW)	$Q_{D(i)}$ (MVar)	PL^0 (kW)	Sum VSI^0 (pu)	VD^0 p.u	Vmin (bus)	Min VSI^0 p.u
Summer Day (Base load) 1.0)	Base	3.5665	1.8597	157.7442	26.53741	0.089689	0.923961 (18)	0.7291340
	1	3.7388	1.9496	174.4589	26.28141	0.099245	0.919997 (18)	0.7167396
	2	3.9194	2.0438	193.0192	26.01372	0.109865	0.915809 (18)	0.7038155
	3	4.1087	2.1425	213.6425	25.73391	0.121676	0.911380 (18)	0.6903430
	4	4.3072	2.2460	236.5747	25.44150	0.134821	0.906695 (18)	0.6763034
Summer Night (Low load) 0.75	5	4.5153	2.3545	262.0946	25.13593	0.149464	0.901735 (18)	0.6616776
	Base	2.6716	1.3416	83.6052	27.93320	0.047202	0.944859 (18)	0.7972432
	1	2.8001	1.4064	92.2836	27.74103	0.052122	0.942049 (18)	0.7878232
	2	2.9359	1.4743	101.8883	27.54003	0.057571	0.939086 (18)	0.7779855
	3	3.0777	1.5455	112.5225	27.32978	0.063607	0.935962 (18)	0.7677138
Winter Day (High load) 1.25	4	3.2264	1.6202	124.3019	27.10988	0.070299	0.932667 (18)	0.7569910
	5	3.3822	1.6985	137.3563	26.87991	0.077719	0.929189 (18)	0.7458000
	Base	4.4038	2.2581	243.2314	25.36686	0.138073	0.905641 (18)	0.6732169
	1	4.6165	2.3671	269.4917	25.05796	0.153087	0.900618 (18)	0.6584633
	2	4.8395	2.4815	298.7442	24.73523	0.169832	0.895296 (18)	0.6431004
Winter Night (Base load) 1.0	3	5.0733	2.6013	331.3609	24.39808	0.188523	0.889653 (18)	0.6271093
	4	5.3184	2.7271	367.7664	24.04594	0.209413	0.883664 (18)	0.6104711
	5	5.5753	2.8587	408.4467	23.67819	0.232787	0.877303 (18)	0.5931672
	Base	3.4987	1.7889	147.4796	26.71120	0.083327	0.926814 (18)	0.7381902
	1	3.6677	1.8753	163.0579	26.46310	0.092177	0.923011 (18)	0.7261855
	2	3.8449	1.9659	180.3474	26.20371	0.102009	0.918994 (18)	0.7136650
	3	4.0306	2.0609	199.5480	25.93252	0.112936	0.914748 (18)	0.7006104
	4	4.2253	2.1604	220.8856	25.64901	0.125091	0.910257 (18)	0.6870031
	5	4.4294	2.2648	244.6153	25.35279	0.138622	0.905506 (18)	0.6728240

daytime after DG integration. Similar trend of system improvement occurs over the planning period.

For Case 2, the installation of 1 FC and 2 ICE improves the performance of the system metrics by reducing the total active power loss to 21.4496113 kW with reference to the base case (i.e., 243.2314 kW) which is equivalent to 91.18 % active power loss reduction. The total VSI in the base year increases from 25.36686 p.u (without DGs) to 31.430219373 p.u (23.91 % improvement) while total VD also reduces from 0.138073 p.u to 0.001168105 p.u (99.15 % reduction). The system voltage profile improves with the minimum voltage increasing to 0.9898434 p.u in the base year in comparison to 0.905641 p.u before the integration of the DGs. In the night when the system is on full load, in the base year, the total active power loss reduces to 12.3168709 kW in comparison to 21.44961134 kW (42.58 % reduction) in the daytime in the base year of operation. Other system parameters are similarly improved as shown in Table A2 for the remaining years of system operation.

In case 3, the integration of 3 FCs DGs had caused reduction in the system's total active power loss, total VSI improvement and decline in total VD of the system in the day and night with respect to respect to the base case (no DG). In the daytime of the base year, the total active respect to the base case (no DG). In the daytime of the base year, the total active power loss of the system reduces from 243.2314 kW to 69.46377866 kW which translates to about 71.44 % reduction. The total VSI of the system increases to 30.23095360 p.u while total VD reduces to 0.008034304 p.u with reference to 25.36686 p.u and 0.138073 p.u, respectively for base case. The system voltage profile also improves with the minimum voltage increasing before the integration of the DGs from 0.905641 p.u to 0.97564253 p.u in bus 29 in the base year. In the winter night, active power loss reduces to 44.85836337 kW in comparison to 69.46377866 kW in the daytime in the first year of operation and 147.4796 kW before integrating DGs. This translates to about 35.42 % reduction when operating the system between the day and the night and about 69.58 % active power loss reduction with reference to base case (i.e., without DGs) in the base year. The total system VSI improves to 30.98342681 p.u when compared with 30.23095360 p.u in the daytime and 26.71120 p.u for base case while VD further reduces to 0.002666469 p.u in the night in comparison to 0.008034304 p.u in the day time. The system voltage profile well improves in comparison to the winter day and the base case.

For Case 4, the integration of 3 ICEs DGs produces a remarkable

improvement on the system performance parameters. The total power loss diminishes from 243.2314 kW in the base case year to 17.31793330 kW with reference to the base case which is equivalent to 92.88 % active power loss reduction. The total VSI in the base year increases from 25.36686 p.u (without DGs) to 31.53556192 p.u (with DGs integration) while total VD also reduces from 0.138073 p.u (without DGs) to 0.000875915 p.u (with DGs). The system voltage profile also improves with the minimum voltage increasing to 0.99140599 p.u in the base year in comparison to 0.905641 p.u in bus 18 before the integration of the DGs. Meanwhile, in the night-time, the total active power loss reduces to 9.88201999 kW in comparison to 17.31793330 kW in the daytime in the base year and 147.4796 kW before integrating the DGs. This translates to about 42.94 % reduction when operating the system between the day and the night and 93.30 % with respect to when no DG was integrated in the night. Other system parameters are similarly improved for the remaining years of system operation both in daytime and at night

Generally, in all cases, optimal integration of FCs and ICEs have remarkably ameliorated the impact of network loading due to load growth with the application of the proposed method. It should be noted that with the load growth, the locations of the DGs remain the same while the capacities are increasing to trail the load growth. Again, it is observed that at night the capacities of the DGs decrease in comparison to the day-time. This will avail the DNO opportunity to remove some DGs from the network and ensure optimal economic dispatch. This action will bring about economic savings in terms of fuel cost reduction.

Moreover, to further investigate the impact of these DG combinations on the system in winter season, the voltage profile and the branch current of each case for fifth year only is presented. This is because the maximum load is connected to the system in this year hence more voltage drop is experienced, and the graphical presentation depicted in Fig. 5. It could be inferred from Fig. 5(a-d) that the system voltage is substantially improved and the branches current drastically reduce with the integration of the DGs to the network both for winter day and night. However, there is more voltage drop and higher line current in the day than night due to increased load demand during the day hours which necessitates for the utilisation of larger DG capacity.

8. Convergence characteristic and statistical analysis

In order to establish the consistency of the proposed IPSO with

reference to the standard PSO, the statistical analysis as well as the convergence characteristics of the objective (fitness) function (OF) values for both algorithms considering all the cases for constant and winter day load models are evaluated. The convergence curves for both methods in base year for constant and winter day seasonal voltage dependent load models are displayed in Fig. 6 and Fig. 7, respectively.

It is apparent from Figs. 6 and 7 that the proposed IPSO has better convergence capability than the PSO, as it reaches a better fitness value in fewer iterations than the standard PSO. This implies that fewer iterations are required for the proposed IPSO in comparison with standard PSO. Due to the stochastic nature of both algorithms, they are subjected to 20 independent runs from which the best, worst, average and standard deviation of the fitness value for each case of DG combination. Tables 10 and 11 depict results of the statistical values obtained by implementing the two methods for constant load and winter day, respectively. In addition, the percentage deviation of the best and worst objective function value from the corresponding mean value are also evaluated using Eqs. 42 and 43 (Doagou-Mojarrad et al., 2013; Jeddi et al., 2019) and the results are shown in Tables 12 and 13, respectively for constant and seasonal voltage dependent load models

$$A = \frac{|Best - Mean|}{Mean} \times 100\% \quad (42)$$

$$B = \frac{|Worst - Mean|}{Mean} \times 100\% \quad (43)$$

Where, A and B are respectively the percentage deviation of the best solution (OF) from the corresponding mean and deviation of the worst solution (OF) from the mean value.

According to Tables 10, 11, 12 and 13, the proposed IPSO algorithm seems to be more efficient than the original PSO, since its best, worst, average and standard deviation as well as the deviations of the mean fitness values for both constant and winter day load models under various DG combinations are lower than those of the original PSO.

9. Conclusion

In this paper, an IPSO algorithm has been applied for optimal placement of B-DGs on a distribution network in order to respond to yearly load growth. The proposed optimisation algorithm finds the

possible best sizes, optimal locations, and power factors of the B-DGs. In this study, different load models including constant and seasonal mixed voltage-dependent load modes and load growth for multi-objective optimisation are considered. The multi-objective problem is solved using weighted sum of three objectives such as total active power loss, voltage deviation, and total voltage stability index. The applicability of the proposed model is tested on a standard IEEE 33 bus system considering different combinations of FC and ICE-DGs for constant and seasonal mixed voltage dependent load models under different loading conditions. The outcomes of the proposed IPSO are compared with the standard PSO as well as recent state-of-the-art optimisation approaches found in literature for only constant load model. The main findings of the results obtained from this paper are:

- It is found that the case study (Soweto) has food waste potential to produce biogas for biogas-based distributed generators.
- The proposed method is able to find optimal sizes, locations and power factor of B-DGs while improving the network performance parameters. It is found that ICE based biogas DG modelled to operate at optimal power factor returns better network performance than unity power factor DGs such as FC for all cases and scenarios considered. To be specific, for constant load model, allocation of 3 ICEs reduces the active power loss to 13.1381 kW (93.52 % loss reduction in comparison with 77.4260 kW (61.80 % loss reduction) for 3 FC-based-DG, 17.0254 kW, 2 ICE and 1FC (91.60 % reduction) and 28.7058 kW for 2FC and 1 ICE (85.84 % reduction) configurations. Similar improved results are obtained for mixed seasonal voltage-dependent load models.
- It is found that yearly load growth causes an increase in the size of the DG but not on the locations. This is quite reasonable as it may be practically and economically unwise to move DGs from one location to another as a result of load growth.
- The use of seasonal mixed load models including daytime and night-time of the season has revealed that lower DG capacities are needed in the night-time compared to the daytime hence, allowing the DNO to disengage some DG units in the night-time thereby reducing the cost of fuel and ensuring optimal economic dispatch of the DGs.
- The proposed IPSO algorithm outperforms the original PSO in terms of quality solution (as shown in Tables 7 and 8), convergence

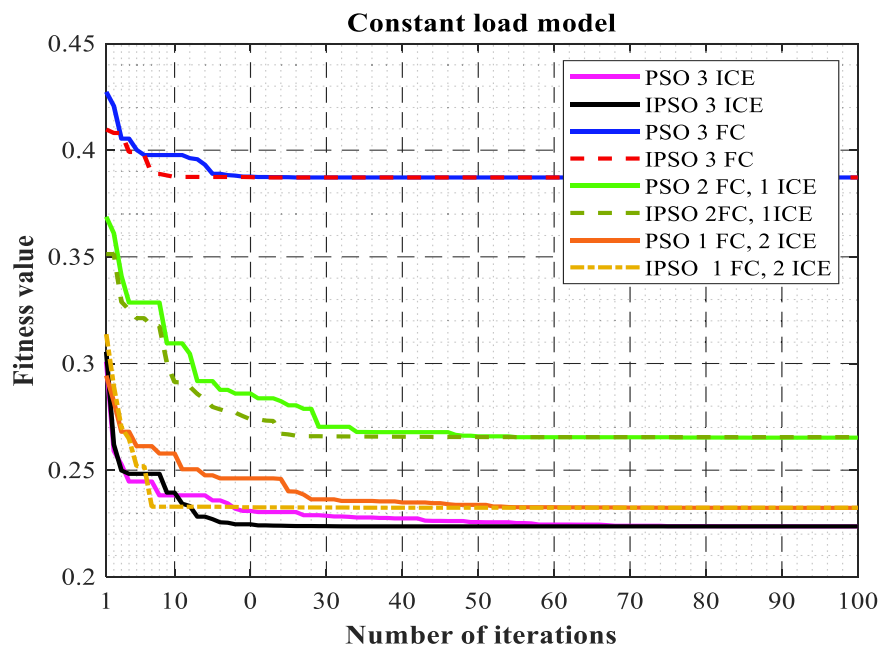


Fig. 6. Comparative convergence curves of the fitness value for IPSO and PSO in constant load model.

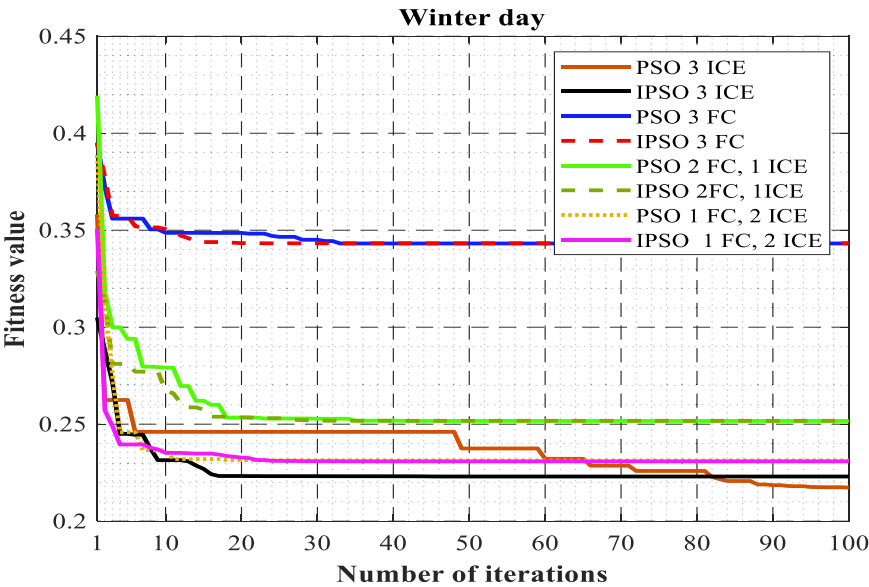


Fig. 7. Comparative convergence curves of the fitness value for IPSO and PSO in seasonal voltage dependent (winter day) load model.

Table 10
Statistical results for both IPSO and PSO for constant load model.

Statistical parameters	PSO				IPSO			
	FC	ICE	2FC, 1ICE	2ICE,1FC	FC	ICE	2FC, 1ICE	2ICE,1FC
Average value	0.395202	0.264455	0.322434	0.256081	0.394382	0.229737	0.283307	0.241209
Minimum value	0.387198	0.223647	0.283251	0.232338	0.387197	0.223645	0.265235	0.232338
Maximum value	0.470659	0.358703	0.469213	0.355148	0.442163	0.309816	0.410798	0.328025
Standard deviation	0.003886	0.005013	0.005313	0.004767	0.002504	0.001697	0.004269	0.004049

Table 11
Statistical results for both IPSO and PSO for seasonal voltage dependent (winter day) load model.

Statistical parameters	PSO				IPSO			
	FC	ICE	2FC, 1ICE	2ICE,1FC	FC	ICE	2FC, 1ICE	2ICE,1FC
Average value	0.359338	0.257140	0.312647	0.279271	0.358558	0.253293	0.303426	0.269899
Minimum value	0.343168	0.217330	0.251770	0.231774	0.343246	0.217178	0.251755	0.230838
Maximum value	0.431575	0.368190	0.419364	0.388702	0.429119	0.353949	0.398877	0.374295
Standard deviation	0.002772	0.006272	0.006095	0.004439	0.001597	0.006054	0.0027746	0.002889

Table 12
Percentage deviation from average for both methods for constant load model.

Deviations from average	PSO				IPSO			
	FC	ICE	2FC, 1ICE	2ICE,1FC	FC	ICE	2FC,1ICE	2ICE, 1FC
A %	2.02529	15.43325	12.15225	9.271676	1.821841	2.651732	6.378954	3.677723
B %	19.09327	35.63858	45.52218	38.68581	12.11541	34.85681	45.35398	35.99202

Table 13
Percentage deviation from average for seasonal voltage dependent (winter day) load model.

Deviations from average	PSO				IPSO			
	FC	ICE	2FC, 1ICE	2ICE,1FC	FC	ICE	2FC,1ICE	2ICE, 1FC
A %	4.4999	15.4818	19.4714	17.0075	4.2704	14.2582	17.0294	14.4725
B %	20.1028	43.1866	34.1334	39.1848	19.6791	39.7389	31.4578	38.6797

capability (see Figs. 6 and 7) and statistical outputs (as shown in Tables 10, 11, 12 and 13) in all the studied cases.

• Based on comparative and functionality assessment with some recent methods, the proposed IPSO is perceived to produce better outcomes in most of the studied cases.

- The practicality of this study could shift the attention of the utilities by using waste as a renewable source of power while also assisting in allocating the right capacity of FC and ICE-based DGs during different period of a season for economic dispatch of the DGs.

As previously mentioned, only technical based objectives are considered in the present study, the future work will incorporate additional objectives such as economic and environmental in order to ensure holistic benefits for all stakeholders in the distributed generation system. A hybrid of the proposed IPSO with other optimisation algorithms such as firefly and cuckoo search for optimal allocation of DGs may be considered for future work. The impact of integration of capacitor bank (CBs) with DGs on optimal allocation could also be investigated in the future. Furthermore, methodology to get Pareto-optimal solutions for the multi-objective allocation of FC and ICE under mixed seasonal voltage dependent load models can be further researched. Larger networks such as IEEE 69 bus and IEEE 118 bus networks could be applied for future work.

CRediT authorship contribution statement

Olawale Mohammed Popoola: Supervision, Writing – review &

editing, Visualization. **Temitope Raphael Ayodele:** Supervision, Visualization, Writing – review & editing. **Moshood Akanni Alao:** Conceptualization, Data curation, Investigation, Methodology, Validation, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgements

This work is based on the research supported wholly / in part by the National Research Foundation of South Africa (Grant Numbers: 150574 & 145900); and Tshwane University of Technology - Faculty of Engineering and Built Environment and Centre for Energy and Electric Power.

Appendix

Table A1
Results for optimal location and capacity of FC and ICE for constant load model

Year	PL^{DG} (kW)	Sum VSI^{DG} (p.u)	VD^{DG} (p.u)	Min(V) (bus)	Min (VSI)	Capacity/location/power factor		
Case 1: Allocation of 2 FC and 1 ICE						FC (kW)/bus/PF	FC (kW)/bus/PF	ICE (kW)/bus/PF
1	28.705831	31.078882	0.00205050	0.988201 (8)	0.9539507	1140.36/24/1.0	932.39/14/1.0	1080.50/30/0.8000
2	31.614767	31.040401	0.00223283	0.987666 (8)	0.9519191	1196.90/24/1.0	979.36/14/1.0	1133.89/30/0.8000
3	34.817413	31.000246	0.00243215	0.987106 (8)	0.9497955	1256.14/24/1.0	1028.61/14/1.0	1190.04/30/0.8000
4	38.350084	30.958377	0.00264993	0.986518 (8)	0.9475763	1318.31/24/1.0	1080.42/14/1.0	1248.92/30/0.8000
5	42.244721	30.914844	0.00288733	0.985904 (8)	0.9452616	1383.55/24/1.0	1134.87/14/1.0	1310.75/30/0.8000
6	46.541958	30.869391	0.00314722	0.985258 (8)	0.9428407	1452.01/24/1.0	1192.14/14/1.0	1375.59/30/0.8000
Case 2: Allocation of 1 FC and 2 ICE						ICE (kW)/bus	ICE (kW)/bus/PF	FC (kW)/bus/PF
1	17.025396	31.631752	0.00055296	0.993777 (25)	0.9754334	836.94/13/0.8502	1080.42/30/0.8000	1137.02/24/1.0
2	18.721231	31.617311	0.00060117	0.993534 (25)	0.9744883	876.91/13/0.8502	1133.89/30/0.8000	1196.90/24/1.0
3	20.554797	31.592946	0.00066153	0.993225 (25)	0.9732882	917.08/13/0.8502	1190.04/30/0.8000	1256.14/24/1.0
4	22.597166	31.574053	0.00072307	0.992915 (25)	0.9720880	965.87/13/0.8502	1248.92/30/0.8000	1318.31/24/1.0
5	25.213569	31.623082	0.00100727	0.992484 (25)	0.9690304	990.58/13/0.8502	1310.75/30/0.8000	1381.88/24/1.0
6	27.283732	31.532230	0.00086747	0.992214 (25)	0.9693707	1052.23/13/0.8502	1375.58/30/0.8000	1450.38/24/1.0
Case 3: Allocation of 3 FC						FC (kW)/bus/PF	FC (kW)/bus/PF	FC (kW)/bus/PF
1	77.426047	30.493037	0.00612932	0.978535 (29)	0.9200119	1032.57/13/1.0	1140.44/24/1.0	1140.44/30/1.0
2	85.397974	30.423938	0.00674944	0.977446 (29)	0.9162543	1085.06/13/1.0	1196.90/24/1.0	1196.90/30/1.0
3	94.206488	30.351648	0.00743521	0.976297 (29)	0.9123137	1140.34/13/1.0	1256.14/24/1.0	1256.14/30/1.0
4	103.94209	30.276019	0.00819403	0.975083 (29)	0.9081818	1198.59/13/1.0	1318.31/24/1.0	1318.31/30/1.0
5	114.70479	30.196846	0.00903472	0.973798 (29)	0.9038455	1259.96/13/1.0	1383.53/24/1.0	1383.53/30/1.0
6	126.46274	30.105161	0.01005915	0.972308 (29)	0.8987931	1324.26/13/1.0	1450.19/24/1.0	1447.47/30/1.0
Case 4: Allocation of 3 ICE						ICE (kW) /bus/PF	ICE (kW)/bus/PF	ICE (kW)/bus/PF
1	13.1381206	31.668805	0.000443206	0.994008 (22)	0.9774548	832.82/13/0.8724	1080.42/24/0.8815	1080.50/30/0.8000
2	14.4334808	31.652666	0.000485572	0.993717 (22)	0.9751125	870.44/13/0.8724	1133.89/24/0.8815	1133.89/30/0.8000
3	15.8591878	31.636723	0.000530683	0.993413 (22)	0.9739212	914.52/13/0.8724	1190.04/24/0.8815	1190.04/30/0.8000
4	17.4001718	31.604793	0.000593976	0.993094 (22)	0.9721348	959.59/13/0.8724	1248.18/24/0.8815	1248.92/30/0.8000
5	19.1464303	31.601633	0.000635341	0.992759 (22)	0.9713606	1003.98/13/0.8724	1310.75/24/0.8815	1310.75/30/0.8000
6	21.0367506	31.582181	0.000696364	0.992407 (22)	0.9699834	1049.19/13/0.8724	1375.52/24/0.8815	1375.59/30/0.8000

Table: A2
Results for optimal location and capacity of FC and ICE for winter day and night of practical mixed load model under load growth and the load conditions

Season	Year	PL^{DG} (kW)	Sum VSI (p.u)	VD (p.u)	Vmin/bus	min VSI	Capacity/location		
Case 1: Allocation of 2 FC and 1 ICE							FC (kW)/loc/PF	FC (kW)/loc/PF	ICE (kW)/loc/PF
Winter day	Base	29.8459809	30.96303313	0.002476436	0.9874869 (7)	0.9509905	1140.44/13/1.0	1140.44/24/1.0	1143.59/30/0.800
	1	32.7845990	30.90643532	0.002757458	0.9868315 (7)	0.9484778	1192.81/13/1.0	1192.38/24/1.0	1199.97/30/0.800

(continued on next page)

Table: A2 (continued)

Season	Year	PL ^{DG} (kW)	Sum VSI (p.u)	VD (p.u)	Vmin/bus	min VSI	Capacity/location		
Winter Night	2	36.1415466	30.87028115	0.002953504	0.9862979 (7)	0.9464428	1256.14/13/1.0	1256.14/24/1.0	1256.89/30/0.800
	3	39.7763205	30.82055551	0.003227583	0.9856586 (7)	0.9440051	1318.31/13/1.0	1318.31/24/1.0	1315.38/30/0.800
	4	43.7769359	30.77478388	0.003493365	0.9850345 (7)	0.9416326	1384.79/13/1.0	1384.79/24/1.0	1374.18/30/0.800
	5	48.1879118	30.72054712	0.003820357	0.9843322 (7)	0.9389673	1453.32/13/1.0	1453.32/24/1.0	1433.28/30/0.800
	Base	17.5369031	31.72381518	0.000544602	0.9940593 (8)	0.9765138	787.45/14/1.0	1076.10/24/1.0	1080.42/30/0.800
	1	19.2856180	31.71058655	0.000601533	0.9937694 (8)	0.9753817	825.23/14/1.0	1127.08/24/1.0	1133.89/30/0.800
	2	21.1762207	31.69075567	0.000666424	0.9934168 (8)	0.9740053	864.57/14/1.0	1180.75/24/1.0	1190.04/30/0.800
	3	23.2055737	31.66237167	0.000747100	0.9929857 (8)	0.9723226	903.89/14/1.0	1236.69/24/1.0	1248.92/30/0.800
	4	25.1378595	31.59287829	0.000810452	0.9928150 (8)	0.9714997	943.19/14/1.0	1294.61/24/1.0	1311.94/30/0.800
	5	27.6523884	31.57227054	0.000783584	0.9924547 (8)	0.9701232	982.59/14/1.0	1354.57/24/1.0	1376.83/30/0.800
Case 2: Allocation of 1 FC and 2 ICE							FC (kW)/loc/PF	ICE (kW)/loc/PF	ICE (kW)/loc/PF
Winter day	1	21.4496113	31.43021937	0.001168105	0.9898434 (25)	0.9600299	1139.64/24/1.0	1080.42/30/0.800	972.62/13/0.9166
	2	22.9709338	31.34953198	0.001337871	0.9893486 (25)	0.9581246	1196.89/24/1.0	1133.89/30/0.800	1059.87/13/0.9166
Winter Night	3	25.2438696	31.32199896	0.001460449	0.9888499 (25)	0.9561996	1256.14/24/1.0	1190.04/30/0.800	1111.59/13/0.9165
	4	27.7439403	31.29328111	0.001594625	0.9883272 (25)	0.9541856	1318.31/24/1.0	1248.92/30/0.800	1165.87/13/0.9165
	5	30.4549576	31.24378991	0.001776841	0.9877379 (25)	0.9519166	1383.54/24/1.0	1310.71/30/0.800	1203.62/13/0.9165
	6	33.4870180	31.23408568	0.001891902	0.9872251 (25)	0.9499514	1453.32/24/1.0	1376.83/30/0.800	1282.0/13/0.9165
	Base	12.3168709	31.68008051	0.000358720	0.9940126 (22)	0.9762715	1063.74/24/1.0	995.09/30/0.800	731.17/13/0.9188
	1	13.5412549	31.66457501	0.000394624	0.9937214 (22)	0.9751284	1114.71/24/1.0	1043.34/30/0.800	766.70/13/0.9187
	2	14.8877852	31.64837648	0.000434074	0.9934159 (22)	0.9739304	1168.17/24/1.0	1093.95/30/0.800	803.97/13/0.9187
	3	16.3687064	31.63145799	0.000477410	0.9930954 (22)	0.9726752	1224.24/24/1.0	1147.02/30/0.800	843.07/13/0.9187
	4	17.9974967	31.61379062	0.000525011	0.9927593 (22)	0.9713598	1283.03/24/1.0	1202.70/30/0.800	884.09/13/0.9187
	5	19.7889967	31.59534459	0.000577293	0.9924066 (22)	0.9699816	1344.69/24/1.0	1261.10/30/0.800	927.12/13/0.9187
Case 3: Allocation of 3 FC							FC (kW) /loc/PF	FC (kW) /loc/PF	FC (kW)/loc/PF
Winter day	Base	69.46377866	30.23095360	0.008034304	0.97564253 (29)	0.9082017	1140.44/13/1.0	1140.44/30/1.0	1140.44/24/1.0
	1	76.57725322	30.14762540	0.008854400	0.97442091 (29)	0.9038842	1196.90/13/1.0	1196.90/30/1.0	1196.90/24/1.0
Winter night	2	84.43102338	30.06008457	0.009763672	0.97313132 (29)	0.8993539	1256.14/13/1.0	1256.14/30/1.0	1256.14/24/1.0
	3	93.10095368	29.96778934	0.010775605	0.97176773 (29)	0.8945936	1318.31/13/1.0	1318.31/30/1.0	1318.31/24/1.0
	4	102.6852536	29.87151983	0.011891813	0.97033142 (29)	0.8896141	1383.55/13/1.0	1383.55/30/1.0	1383.55/24/1.0
	5	113.2714661	29.76999402	0.013135594	0.968812 (29)	0.8843808	1452.01/13/1.0	1452.01/30/1.0	1452.01/24/1.0
	Base	44.85836337	30.98342681	0.002666469	0.984914 (33)	0.9430446	894.60/13/1.0	1140.44/30/1.0	1137.62/24/1.0
	1	49.33216202	30.92736144	0.002977497	0.984116 (33)	0.9401975	935.43/13/1.0	1196.90/30/1.0	1190.57/24/1.0
	2	54.26260060	30.86859921	0.003323644	0.983275 (33)	0.9372133	978.31/13/1.0	1256.14/30/1.0	1246.20/24/1.0
	3	59.69745289	30.80701797	0.003708830	0.982388 (33)	0.9340863	1023.32/13/1.0	1318.31/30/1.0	1304.64/24/1.0
	4	65.68940653	30.74247234	0.004137556	0.981453 (33)	0.9308081	1070.59/13/1.0	1383.55/30/1.0	1366.06/24/1.0
	5	72.29711731	30.67482858	0.004614719	0.980466 (33)	0.9273727	1120.23/13/1.0	1452.01/30/1.0	1430.60/24/1.0
Case 4: Allocation of 3 ICE							ICE (kW)/loc/PF	ICE (kW)/loc/PF	ICE (kW)/loc/PF
Winter day	Base	17.31793330	31.53556192	0.000875915	0.99140599 (33)	0.9660695	1042.05/13/0.9309	1080.42/24/0.8637	1080.42/30/0.800
	1	19.03921724	31.52026945	0.000952853	0.99106762 (33)	0.9647518	1093.86/13/0.9309	1133.89/24/0.8637	1133.89/30/0.800
Winter Night	2	20.93255210	31.50469133	0.001036624	0.99071744 (33)	0.9633895	1148.28/13/0.9309	1190.04/24/0.8637	1190.04/30/0.800
	3	23.01662002	31.48873562	0.001128243	0.99035219 (33)	0.9619702	1205.48/13/0.9309	1248.92/24/0.8637	1248.92/30/0.800
	4	25.30924462	31.47268638	0.001227823	0.98997630 (33)	0.9605112	1265.61/13/0.9309	1310.75/24/0.8637	1310.75/30/0.800
	5	27.83053645	31.45574743	0.001337586	0.9895805 (33)	0.9589765	1328.69/13/0.9309	1375.59/24/0.8637	1375.59/30/0.800
	Base	9.88201999	31.71500641	0.00029919	0.994125 (22)	0.9767165	738.28/13/0.9342	1058.07/24/0.9158	974.14/30/0.800
	1	10.8630112	31.70113164	0.000329083	0.993839 (22)	0.9755946	774.13/13/0.9342	1108.76/24/0.9158	1021.36/30/0.800
	2	11.9416798	31.68663869	0.000361907	0.993539 (22)	0.9744189	811.73/13/0.9342	1161.92/24/0.9158	1070.87/30/0.800
	3	13.1277899	31.67150420	0.000397951	0.993225 (22)	0.9731870	851.18/13/0.9342	1217.67/24/0.9158	1122.81/30/0.800
	4	14.4320840	31.65570267	0.000437527	0.992896 (22)	0.9718962	892.55/13/0.9342	1276.13/24/0.9158	1177.29/30/0.800
	5	15.8663849	31.63920765	0.000480977	0.992549 (22)	0.9705436	935.95/13/0.9342	1337.44/24/0.9158	1234.43/30/0.800

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